

Genetic Distance between Two Gene Loci and Recombination Frequency

The closer together two gene loci are located, the more frequently they are inherited together (genetic linkage); the farther apart, the more frequently they become separated by recombination. The highest possible frequency of recombination is 50% (0.50), because this corresponds to the frequency of segregation of genes on different chromosomes. Thus, the frequency of recombination reflects the distance between two loci (genetic distance). This distance can be expressed as the frequency of genetic recombination (as opposed to the physical distance, which is given as the number of DNA base pairs lying between the two loci, see p. 240).

Syteny (H. J. Renwick, 1971) refers to gene loci being located on the same chromosome, whether or not they are linked. Thus the term syteny also includes unlinked, widely separated loci on the same chromosome.

A. Recombination frequency as a consequence of the distance between two loci

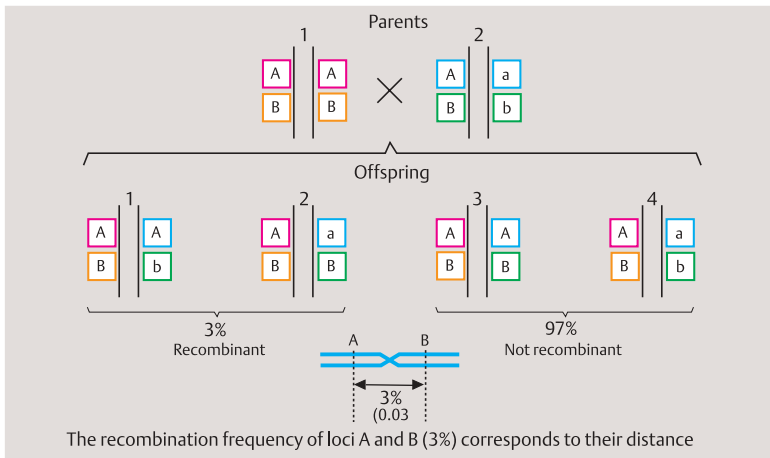
Two neighboring gene loci **A** and **B** in the parents may either become recombinant or remain nonrecombinant (see p. 144). If one of the parents is heterozygous for two alleles **Aa** and **Bb**, but the other homozygous for both, then homozygosity at only locus **A** (1) or only locus **B** (2) in the offspring will be the result of recombination. The observed recombination frequency between locus **A** and locus **B** (3%) results from the distance between them. These two loci are said to be 0.03 recombination units (3 cM) apart. One recombination unit is a centimorgan (cM), and 1 cM corresponds to a recombination frequency of 1% (0.01). In mammals, recombination occurs more often in female meiosis than in male meiosis, so that the genetic distance in females is about 1.5 times greater than in males (see p. 240). The term morgan is derived from the name of the American geneticist who in 1911 first described recombination in *Drosophila*. At that time, the observation of linkage and recombination was an important argument for genes being linearly arranged along the chromosomes.

B. Determination of the order of three gene loci and their relative distances from each other by measuring recombination frequency

Not only the relative distances between gene loci but also their order can be determined by comparing recombination frequencies. In the example presented, the order of three gene loci, **A**, **B**, and **C** of unknown distance from each other is to be determined (1). In plants and animals, the distance between any two of the loci (locus **A** from locus **B**, locus **B** from locus **C**, locus **A** from locus **C**) can be established.

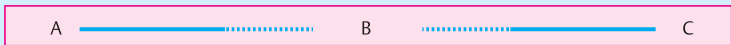
In traditional experimental genetics, such hybridization experiments were used for this purpose. For a test cross, differing parental genotypes can be used. One parental genotype is crossed with a different one. The observed recombination frequencies (2) indicate the relative distances of the loci from each other. In the example presented, the distance from locus **A** to locus **C** is 0.08 (8%); the distance between locus **B** and locus **C** is 0.23 (23%); and the distance between **A** and **B** is 0.31 (31%). Thus, with the distance 0.08 between **A** and **C** and 0.23 between **C** and **B**, they are located in the order **A–C–B** (3).

This type of indirect determination of the relative location of gene loci in their correct order in classical experimental genetics has been replaced by direct methods of recombinant DNA technology.

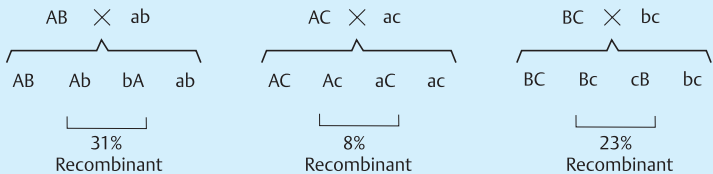


A. Recombination frequency as a consequence of the distance of two loci

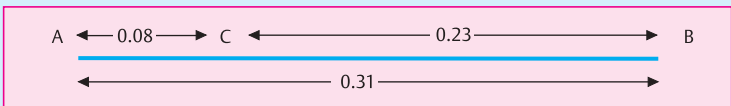
1. Gene loci A, B, C of unknown distance



2. Test cross of homozygous parental genotypes



3. Relative distance



B. Determination of the relative distance and sequence of three gene loci by measuring the frequency of recombination

Segregation Analysis with Genetic Markers

Individual differences in the sequence of DNA nucleotide bases (DNA polymorphism) can be used as genetic markers to obtain indirect information about the presence or absence of a mutation at a closely linked disease locus (indirect gene diagnosis). Segregation analysis of linked markers and a disease locus can be applied even when a mutation cannot be found. By using the known linkage between a DNA marker and a disease locus, important information can be obtained about the possible occurrence of a genetic disease. Three examples of segregation analysis in each of the three modes of monogenic inheritance are given below. A broad repertoire of genetic markers with high information content is available (see microsatellites and minisatellites, p. 72).

A. Autosomal dominant

Two pedigrees are presented that were studied by marker analysis, one without recombination between disease locus and marker (1), one with recombination (2). Above, the pedigree is shown; below, a diagram of a Southern blot analysis or some other marker system that distinguishes two alleles 1 and 2. A given individual will be homozygous (1-1, or 2-2) or heterozygous (2-1) at this locus (marker locus). In pedigree 1, without recombination, the mother (individual 2) and two children (individuals 4 and 5) are affected with an autosomal dominant disorder. The mother is heterozygous 1-2 for the two DNA markers. The father is homozygous 2-2 for allele 2. All affected individuals in this pedigree carry allele 1, all unaffected individuals do not. Thus, allele 1 must carry the mutation. In family 2, recombination has occurred in individual 5. The frequency of recombination determines the frequency of false predictions. For this reason, very closely linked markers with very low recombination frequencies are used, preferably marker loci that flank the disease locus on both sides.

B. Autosomal recessive

The two affected individuals (individuals 4 and 7) in the left-hand pedigree are homozygous for allele 1 (1-1), inherited from each of their parents. Thus one allele 1 of the father and allele

1 of the mother represent the allele that carries the mutation at the closely linked disease locus. The unaffected sibs (individuals 3, 5, and 6) have received allele 2 from their mother and allele 1 from their father. Since allele 2 does not occur in the affected, it cannot carry the mutation. In this case it cannot be determined whether the unaffected children are heterozygous for the paternal mutant allele 1 or not.

In the pedigree on the right, recombination must have occurred because this child is not affected, although it is homozygous for allele 2 (2-2) as are the affected sibs (individuals 3 and 5).

C. X chromosomal

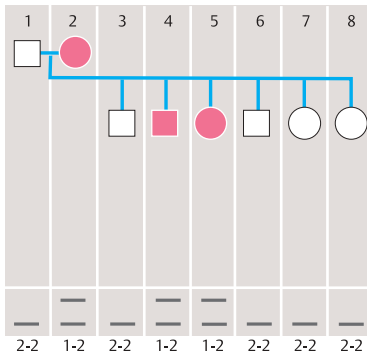
Here the segregation of an X-linked disease is shown in relation to the segregation of a linked marker locus represented by alleles 1 and 2. Three affected male individuals (5, 8, 10) are hemizygous for allele 1. Thus, this allele represents the mutation. All unaffected male individuals are hemizygous for allele 2. The female individuals 2 and 6 are heterozygous 1-2. They must be considered obligate heterozygotes for the mutation. Recombination must have occurred in individual 13 because he is hemizygous for allele 2 but is affected.

The examples show how information about the presence or absence of a mutation can be derived from segregation analysis of closely linked markers (indirect DNA diagnosis).

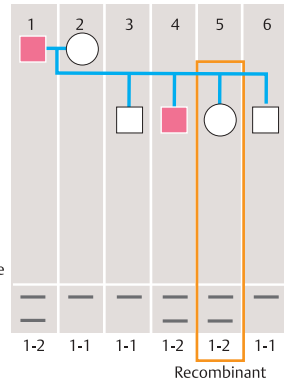
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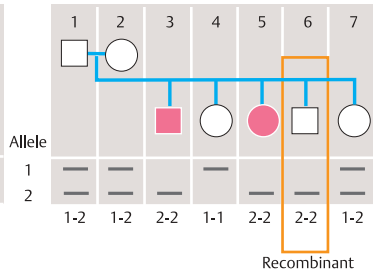
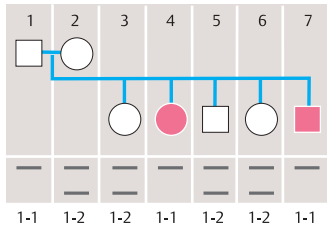
1. Without recombination



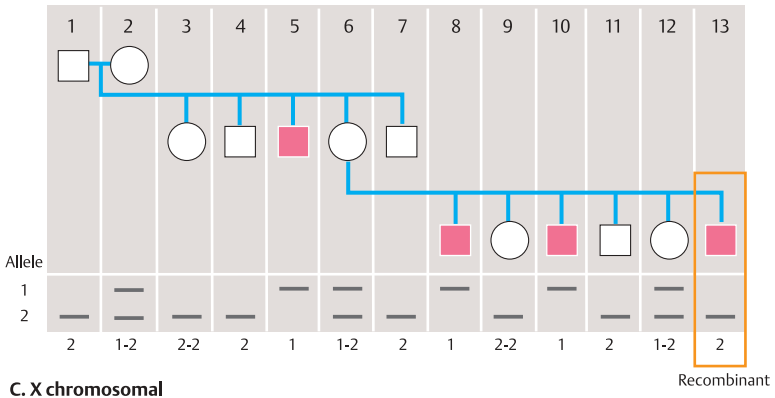
2. With recombination



A. Autosomal dominant



B. Autosomal recessive



C. X chromosomal

Linkage Analysis

Linkage analysis is a test designed to detect the inheritance pattern of two or more gene loci located sufficiently close to each other on the same chromosome. The alleles present on linked loci will tend to be inherited together if recombination between them is rare. The frequency of recombination is a measure of their distance (see p. 146). The analysis of linkage rests on many different sophisticated methods that are supported by computer programs. Here a simplified outline of the background is provided. Three situations are frequently encountered: (i) linkage analysis of two loci, e.g., a locus of interest, mutations of which cause a disease (disease locus) and a locus characterized by a detectable DNA polymorphism (marker locus); (ii) linkage analysis of several loci (multilocus analysis); and (iii) linkage analysis involving the entire genome by means of DNA markers (microsatellites) along each chromosome (genome scan). Each requires different procedures and rests on different assumptions.

A. LOD scores

Linkage of two gene loci is assumed when the probability for linkage as opposed to the probability against linkage is equal to or greater than the ratio of 1000:1 ($10^3:1$). The logarithm of this ratio (odds) is called the LOD score (logarithm of the odds). A LOD score of 3 corresponds to an odds ratio of 1000:1. The likelihood of linkage as opposed to nonlinkage for different recombination fractions (recombination frequencies), expressed as LOD scores, is determined by observations in families. The closer two loci lie to each other, the higher the resulting LOD score. The table shows (in a simplified manner) the LOD score for close linkage with a recombination fraction under 0.05 (a), high probability of linkage with a recombination fraction of 0.15 (b), weak linkage (c), and no linkage (d). If the LOD score is less than 0, linkage is excluded (not shown in the table).

B. LOD score curves for different recombination fractions

The diagram shows a simplified form of the LOD score curves for the values in the table in A. (Figure after Emery, 1986).

C. Multilocus analysis

Because of the great number of available markers at the DNA level, linkage analysis today is usually carried out with multiple markers (multilocus analysis). With the chromosomal position of the marker taken into consideration, the localization score is determined as the logarithm of the probability quotient (likelihood ratio). The localization score of the locus being sought is noted with respect to each of the marker loci (A, B, C, D). Each of the four peaks expresses linkage. The highest peak marks the probable location of the gene being sought. If there is no peak, linkage is not present, and the locus being studied does not map to the region tested (exclusion mapping).

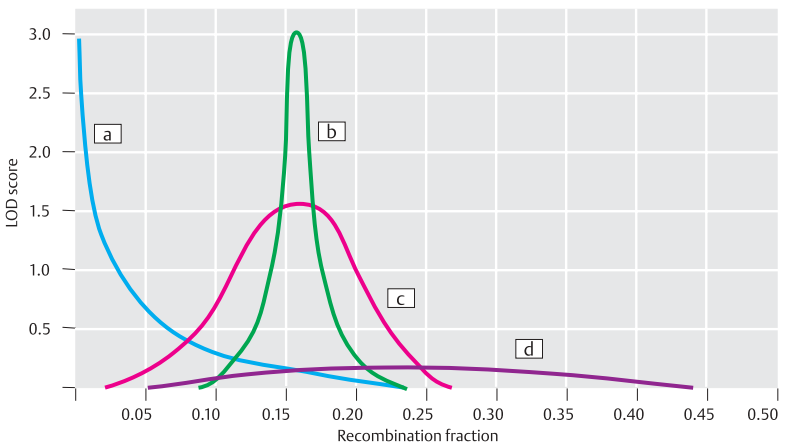
In contrast to linkage, which refers to the genetic distance of gene loci, the term association refers to the co-occurrence of alleles or phenotypes. If one particular allele occurs more frequently in people with a disease than expected from the individual frequencies, this is called an association. There may be many reasons for an association, including nongenetic causes.

References

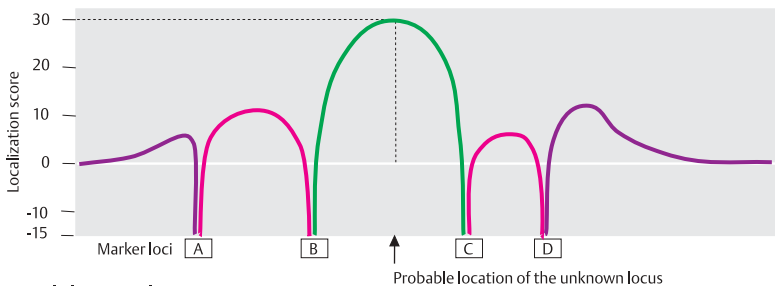
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	Recombination fraction									
	<0.05	0.05	0.10	0.15	0.20	0.30	0.35	0.40	0.45	0.50
a	3	0.7	0.3	0.2	0.01	0				
b		0	0.1	3	0.2	0				
c	0	0.2	0.7	1.6	1.0	0				
d		0	0.1	0.2	0.3	0.2	0.1	0.1	0	

A. LOD scores



B. LOD score curves for close (a), probable (b), and no linkage (c,d)



C. Multilocus analysis

Quantitative Differences in Genetic Traits

Most variation among organisms is not qualitative but quantitative. Height, weight, shades of color in flowers or skin, behavioral patterns, learning capacity, metabolic rate, and blood pressure are examples. They cannot be grouped into separate, easily distinguished categories as can monogenic traits displaying a Mendelian mode of inheritance. The term quantitative genetics was introduced in 1883 by Francis Galton to refer to the variation of complex traits. R. A. Fisher in 1918 and other researchers later laid the foundations to assess the genetics of such traits. Distinguishing genetic and environmental contributions to the trait is a primary goal. As this is usually not possible for individuals, quantitative traits have to be defined for populations. This poses considerable difficulties in defining a trait (question of cut-off point for a trait to be considered abnormal) and finding an appropriate control (comparison with another population). The underlying genetic variation (genotypic variation) is interchangeably called polygenic (many genes), multigenic (several genes), or multifactorial (many factors including nongenetic contributions to the variation). It is important to understand that, at least in theory, a quantitative trait is the result of distinct genotypes at each of the gene loci involved.

A. Length of the corolla in *Nicotiana longiflora*

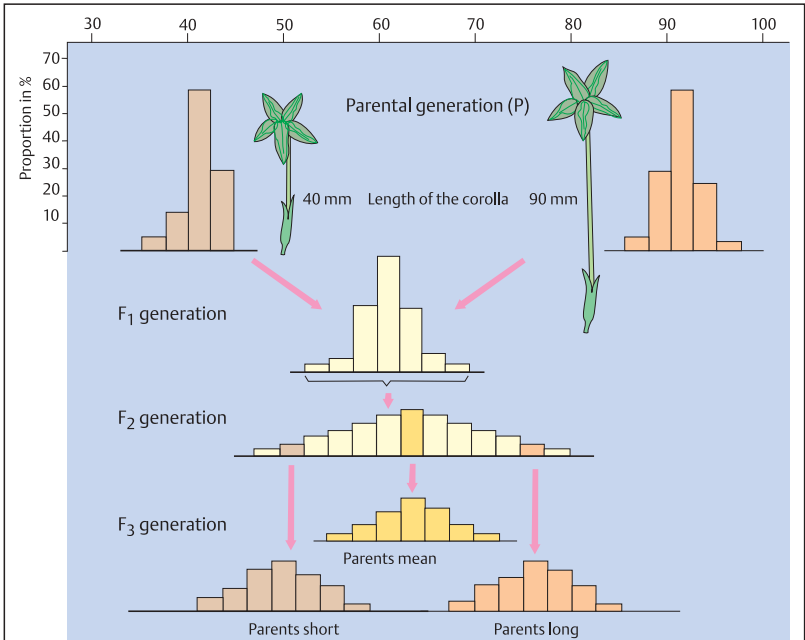
When parent plants with average corolla lengths of 40 cm and 90 cm are crossed, the next generation (F_1) shows corolla length distribution that is longer than that of the short parent and shorter than that of the tall parent. In the next generation (F_2) the distribution spreads at both ends towards long and short. If plants from the short, middle, and long varieties are used (crossed to breed a new population, F_3), the mean resulting distribution corresponds to the corolla length of the parental plants, i.e., short (shown on the left), long (shown on the right), or average (shown in the middle). This can be explained by a difference in the distribution of genes contributing to the variation in the trait, in this example length of the corolla. (Figure adapted from Ayala and Kiger, 1984).

B. Distribution of frequency in the F_2 generation with different numbers of involved gene loci

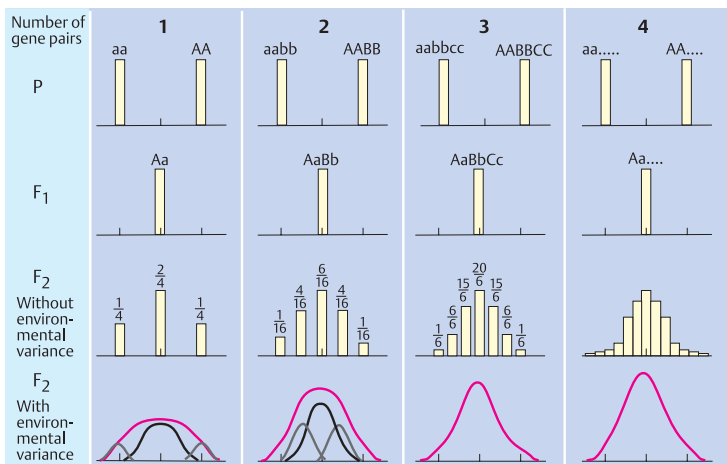
A variable characteristic due to different alleles at a single gene locus can be defined by observable differences in the two types of homozygotes **aa** and **AA** (column 1 in the figure). As the number of loci contributing to the phenotype increases from two (column 2) to three (column 3) or to many loci (column 4), the F_1 generation tends to deviate towards a mean. In the next generation (F_2) the distribution of the frequency of occurrence of the trait spreads. It is no longer possible to use the phenotype to infer the genotype. As the number of loci increases, a smooth distribution of the quantitative trait in the population becomes apparent. Even with just three loci contributing to the phenotype (column 3), the distribution curve is relatively smooth, fitting into a Gaussian (normal) curve as seen with many loci (column 4). Each individual's phenotype is the result of the contribution of each allele at the loci involved (their identity generally remains unknown). The variance of the phenotype (V_P) is the sum of the genetic variance (V_G) and the environmental variance (V_E). However, in many cases these types of variance cannot be distinguished, especially in humans. A locus contributing to a quantitative characteristic is called a quantitative trait locus (QTL). There are many approaches to assessing QTLs (see *Trends Genet.* 11: 463–524, 1995 Special Issue).

References

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A. Length of the flower of *Nicotiana longiflora* after a cross between different types of parental plants



B. Distribution of frequency in the F₂ generation with a different number of gene loci

Normal Distribution and Polygenic Threshold Model

Polygenic inheritance of quantitative traits occurs in all animal and plant species. The analysis of polygenic inheritance requires the application of statistical methods. These aid in assessing the difference between a sample of measurements and the population from which they are derived and define how much confidence can be placed in the conclusions.

A. Normal distribution of a quantitative trait

When quantitative data from a large sample are plotted along the abscissa and the number of individuals along the ordinate, the resulting frequency distribution assumes a bell-shaped Gaussian curve. The mean ($\bar{x}$) intersects the curve at its highest point and divides the area under the curve into two equal parts (1). Further perpendicular intersections can be placed one standard deviation (s) to the left ($-1s$) and to the right ($+1s$) to yield two additional areas, c to the left (e.g., low blood pressure or low height) and d to the right (e.g., high blood pressure or increased height). Areas a and b each comprise 34.13% of the total area under the curve (2). Further partitioning with perpendiculars to the abscissa at two and three standard deviations ($-2s$ and $-3s$ to the left and $+2s$ and $+3s$ to the right) results in further subsections (3).

The mean ($\bar{x}$) of a sample is determined by the sum of individual measurements ($\sum x$) divided by the number of individuals (n) (Formula 1). Many measurements yield the frequency f_x of observed individuals (Formula 2). The population variance (σ^2) defines the variability of the population (Formula 3). It is expressed as the square of the sum ($\sum$) of individual measurements (x) minus the population mean (μ), divided by the number of individuals in the population (N) (Formula 3). However, the variance of the population usually cannot be determined directly. Therefore, the variance of the sample (s^2) must be estimated (Formula 4). The square of the sum of the individual measurements (x) minus the mean ($\bar{x}$) is divided by the number of measurements (n). A correction factor $n/n-1$ is introduced because the number of independent measurements is $n-1$, resulting in

a simplified formula 5. The standard deviation (s , formula 6), the square root of the sample variance (s^2), rests on a large number of successive samples, which in practice might be difficult to obtain.

B. Polygenic threshold model

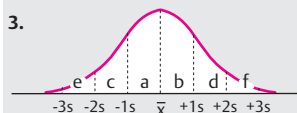
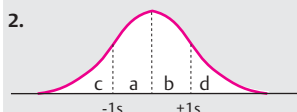
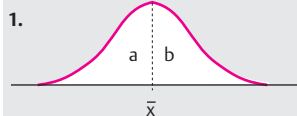
Some continuously variable characters are assumed to represent a susceptibility to a disease. Therefore, it was postulated that if susceptibility exceeds a threshold, the disease is manifest as, for example, shown in B for cleft lip/cleft palate (Fraser, 1980). According to the threshold model, populations differ with respect to the susceptibility to the disease (1). The liability to disease in the offspring of affected individuals (first-degree relative, 2) or in second-degree relatives (3) is closer to the threshold than in unrelated individuals from the general population. In general, the liability of first-degree relatives is about half that of the general population ($1/2 \bar{x}$) and about one quarter ($1/4 \bar{x}$) for second-degree relatives.

C. Different thresholds

Instead of a difference of the distribution of liability for a polygenic trait, the threshold may differ. This is the case when there is a sex difference in the frequency of the disease (1). For example, if the disease occurs more frequently in males than in females, then the susceptibility to the disease is assumed to be greater in males owing to a lower sex-specific threshold. Accordingly, the proportion of affected persons among the offspring of affected males is lower than in the offspring of females (2). Consequently, the proportion of affected offspring of females exceeds that of males (3).

References

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$$1 \quad \bar{x} = \frac{\sum x}{n} \quad (\text{Mean})$$

$$2 \quad \bar{x} = \frac{\sum fx}{n}$$

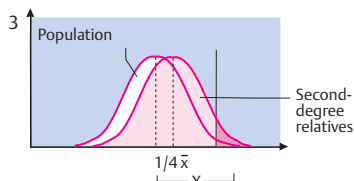
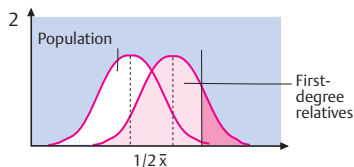
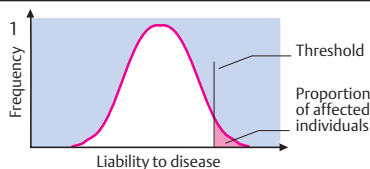
$$3 \quad \sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad \text{Population variance}$$

$$4 \quad s^2 = \left(\frac{\sum (x - \bar{x})^2}{n} \right) \left(\frac{n}{n-1} \right)$$

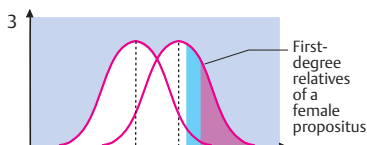
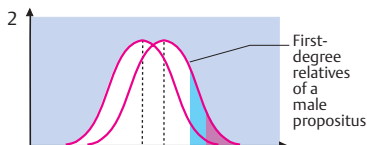
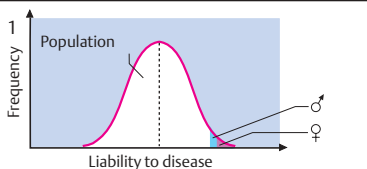
$$5 \quad s^2 = \frac{\sum (x - \bar{x})^2}{(n-1)}$$

$$6 \quad s = \sqrt{\frac{\sum f(x - \bar{x})^2}{n-1}} \quad (\text{Standard deviation})$$

A. Normal distribution of a trait



B. Polygenic threshold model



C. Different thresholds

Distribution of Genes in a Population

A population can be characterized by its distribution of genes, i.e., the frequency of alleles at different gene loci. The field of population genetics is concerned with the distribution patterns of alleles and the factors that alter or maintain their frequencies. A detrimental allele that causes disease in an individual might be expected to disappear sooner or later from a population. For the reasons given below, this is usually not the case. The frequency of an allele in a population is related to the distribution of genotypes among the offspring of different parental genotypes.

A. Expected frequency of genotypes in the children of parents with various genotypes

As shown on p. 140, a gene locus with two alleles (a dominantly inherited allele **A** and a recessively inherited allele **a**) can show six possible combinations (1–6). Each of these has an expected distribution of genotypes in their offspring according to the Mendelian laws. If one parent is homozygous and the other heterozygous, as in combinations 2 and 4, a 1:1 distribution of genotypes will result in the offspring (0.5 **AA** and 0.5 **Aa**). If both parents are heterozygous (combination 3), three genotypes will occur among their children in a ratio of 1:2:1 (0.25 **AA**, 0.5 **Aa**, and 0.25 **aa**). With combinations 1, 5, and 6 the respective offspring are of one genotype.

B. Allele frequency

Population genetics is concerned with the frequencies of different genotypes in a population. The frequency with which an allele is present at a given gene locus in a given population is called allele frequency (or also, gene frequency). The concept of allele frequency refers exclusively to the frequency of the allele in a population, and not directly to the frequency of the individual genotypes.

For a gene locus with two possible alleles **A** and **a**, the only possible genotypes are **AA** or **Aa** or **aa**. The frequency of the two alleles together (p the frequency of **A** and q the frequency of **a**) must be 100% (1.0). If two alleles **A** and **a** are equally frequent (each 0.5), they have the

frequency of $p=0.5$ for the allele **A** and $q=0.5$ for the allele **a** (1); i.e., $p+q=1$. The frequency distribution of the two alleles in a population follows a simple binomial relationship: $(p+q)^2=1$. Accordingly, the distribution of genotypes in the population corresponds to $p^2+2pq+q^2=1.0$. The expression p^2 corresponds to the frequency of the genotype **AA**; the expression $2pq$ corresponds to the frequency of the heterozygotes **Aa**; and q^2 corresponds to the frequency of the homozygotes **aa**.

When the frequency of an allele is known, the frequency of the genotype in the population can be determined. For instance, if the frequency p of allele **A** is 0.6 (60%), then the frequency q of allele **a** is 0.4 (40%). Thus, the frequency of the genotype **AA** is 0.36; of **Aa** it is $2 \times 0.24 = 0.48$; and of **aa** it is 0.16 (2). And conversely, if genotype frequency has been observed, the allele frequency can be determined. If only the homozygotes **aa** are known (e.g., they can be identified owing to an autosomal recessive inherited disease), then q^2 corresponds to the frequency of the disorder. From $p=1-q$, the frequency of heterozygotes ($2pq$) and of normal homozygotes (p^2) can also be determined.

		Genotype	
		of parents	of offspring
1		AA and AA	1.0 AA
2		AA and Aa	0.50 AA 0.50 Aa
3		Aa and Aa	0.25 AA 0.50 Aa 0.25 aa
4		Aa and aa	0.50 Aa 0.50 aa
5		AA and aa	1.0 Aa
6		aa and aa	1.0 aa

A. Expected frequency of genotypes in children of parents with different genotypes

	Parents	
	0.5 A	0.5 a
0.5 A	AA 0.25	Aa 0.25
0.5 a	Aa 0.25	aa 0.25
	Offspring	

$$p = 0.50 \text{ (Frequency of A)}$$

$$q = 0.50 \text{ (Frequency of a)}$$

	A = 0.60 a = 0.40	
	A 0.6	a 0.4
A 0.6	0.36 AA	0.24 Aa
a 0.4	0.24 Aa	0.16 aa
	p	q

$$p^2 + 2pq + q^2 = 1$$

$$0.36 + 0.48 + 0.16 = 1.0$$

$$(AA) \quad (Aa) \quad (aa)$$

B. Allele frequency

Hardy-Weinberg Equilibrium

When an allele in the homozygous or in the heterozygous state leads to a disorder that owing to its severity and the time of onset prevents the individual from reproducing, it represents a selective disadvantage for that individual. Although it could be assumed that such a mutant allele would eventually disappear from a population, this is not the case. Rather, a frequency equilibrium develops according to principles and conditions elucidated in 1908 by the English mathematician G. F. Hardy and the German physician W. Weinberg (Hardy-Weinberg equilibrium principle).

A. Constant allele frequency (Hardy-Weinberg equilibrium)

An allele (here referred to as allele **a**) that leads to a severe disorder in the homozygous state is mainly present in a population in the undetectable heterozygous state. Only homozygotes (**aa**) come to attention as a result of illness. The heterozygotes (**Aa**) remain unrecognized and can not be distinguished by phenotype from the homozygotes **AA**. The frequency of affected individuals (homozygotes **aa**) is determined by the frequency of allele **a** (corresponding to q). The frequency of the three genotypes is determined by the binomial relationship $(p + q)^2 = 1$, where p represents the frequency of allele **A**, and q the frequency of allele **a**. The homozygous alleles (**aa**) eliminated in one generation by illness are replaced by new mutations. An equilibrium is reached between the elimination due to illness and the mutation frequency.

B. Some factors influencing the allele frequency

The Hardy-Weinberg equilibrium principle is valid only under certain conditions: Above all, it applies only when the population is sufficiently large and mating is random (panmixia). Selection for a particular allele would lead to an increase in the corresponding allele frequency. A change in mutation frequency can change the frequency of an allele. However, after a single alteration of the mutation frequency, equilibrium will occur at a new level within one generation.

In a small population, the proportion of genotypes may be shifted owing to casual fluctuations (genetic drift). An allele that is very rare or

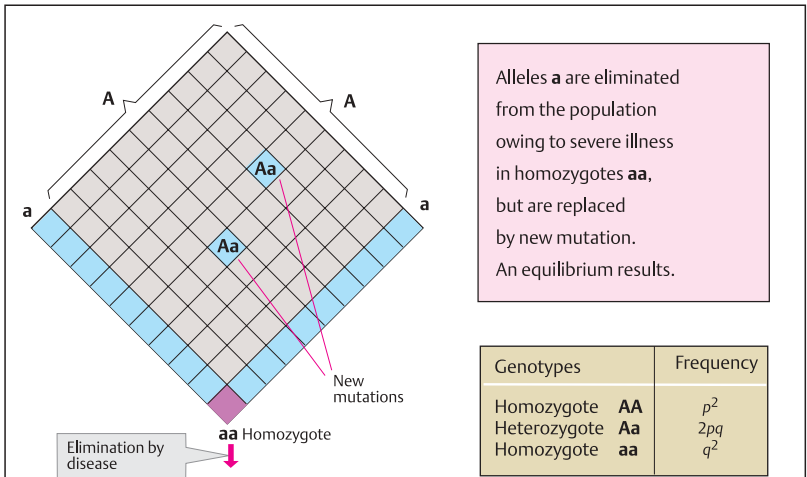
not present in a population can be introduced by migration and spread out as the population grows (founder effect).

Finally, a population may experience a drastic reduction in size, followed by an increase in the number of individuals. An allele that was previously rare in this population may by chance subsequently become relatively common as the population expands again.

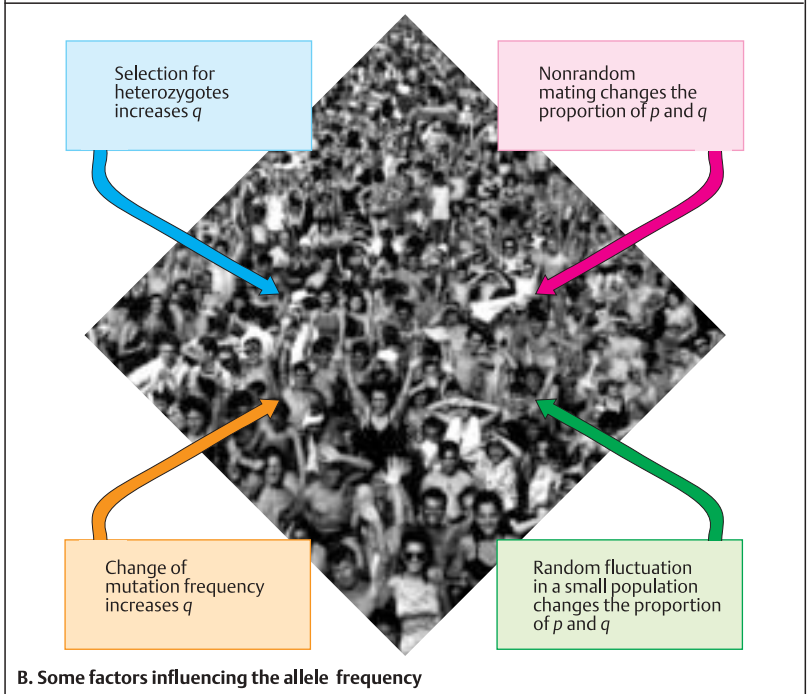
Linkage disequilibrium is a deviation of the frequency of the co-occurrence of certain alleles at two or more neighboring (linked) loci from that expected from their individual frequencies in the population. Linkage disequilibrium may result from a selective advantage of one allele or from a geographic difference with respect to the time of occurrence of a mutation in one population compared to another. In such a case one finds a disease-causing mutation preferably on one haplotype of closely linked marker loci, but rarely or not at all on another (see hemoglobin disorders as an example). (Photographic detail from "Coney Island, 1938, photograph by Weegee," hilma hilcox, Fotofolio).

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A. Constant allele frequency (Hardy Weinberg equilibrium)



Consanguinity and Inbreeding

Consanguinity (being “of the same blood”) of the parents of an individual (parental consanguinity) means that they are genetically related because they have one or more ancestors in common. Consanguinity increases the probability that two related individuals carry an identical mutation as a recessive allele inherited from a common ancestor.

A. Simple types of consanguinity

A mating between closely related individuals is called inbreeding. The degree of inbreeding depends on the closeness of their relationship. A mating between brother and sister or father and daughter is called incest (1). Sibs share half their genes, corresponding to a coefficient of relatedness of $1/2$. The chance of homozygosity by descent at a given locus in their offspring is $1/4$ ($1/2$ each from C and D to E). Two of the most frequent types of consanguineous unions in man are of first (2) and second cousins. First cousins share $1/8$ of their genes. The chance of homozygosity by descent in their offspring is $1/16$. An uncle–niece union is another example of consanguinity (3). They share $1/4$ of their genes. The chance of homozygosity by descent in their offspring is $1/8$. The degree of relationship of two individuals is expressed as coefficient of relationship r . The value of r corresponds to the proportion of genes that are identical by descent in the two persons ($1/2$ for sibs, $1/8$ for first cousins, $1/32$ for second cousins, $1/4$ for aunt–nephew or uncle–niece). The (generally) remote possibility that the unrelated individuals E or F transmit a mutant (allozygous, see below) allele at this locus can usually be disregarded.

B. Identity by descent (IBD)

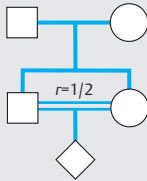
If related individuals mate, the chance increases that a particular allele present in one of their ancestors will have been inherited by both individuals (identity by descent, IBD). This results in an extra chance that their offspring will be homozygous for this allele (autozygous since the alleles are identical by descent, in contrast to allozygous when the two alleles are not identical by descent). The resulting probability of homozygosity at a given locus is the inbreeding coefficient, F .

The pedigree in panel B shows a consanguineous mating between individuals III and IV

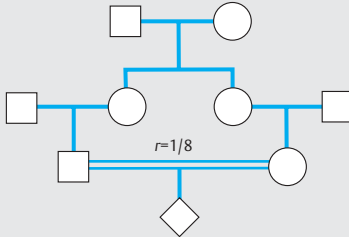
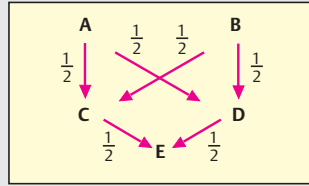
(first cousins) who share two common ancestors, I and II (their grandparents). The risk for an autosomal recessive disease in their offspring V due to homozygosity for a mutant allele at a given gene locus depends on the probability that both parents (III and IV) are heterozygous for this allele. This probability depends on the genetic distance from their common ancestor and the frequency of the allele in the population. In this case, the path of one and the same allele **A** present in one common ancestor I to individual III consists of two generational steps. Since the probability of transmission at each step (equivalent to one generation) is one half (50%), the probability that transmission occurred at both steps is $1/2 \times 1/2$, or $1/4$. However, homozygosity for the allele **A** in individual V requires that individual IV also has inherited allele **A** from the same common ancestor (I) as individual III. The number of steps required is also $1/2 \times 1/2$. The joint probability that allele **A** has descended from individual I to both individuals III and IV, therefore, is $(1/2)^4$ or $1/16$. Since there are two common ancestors and we do not know from which ancestor the allele in common to III and IV has descended (the common allele could also be **C** or **D**), we have to add this possibility to the path from I to III and IV. Therefore, for both paths possible we have $(1/2)^4 + (1/2)^4$ or $1/8$. The final probability of homozygosity of V resulting from identity by descent (IBD) is one half of the proportion of parental genes that are identical by descent ($1/16$).

References

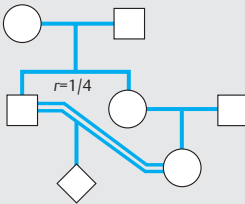
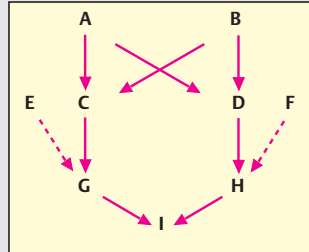
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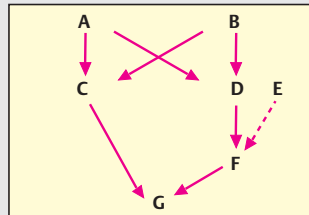
1. Brother/sister mating



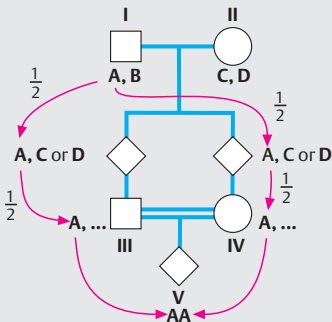
2. First cousins



3. Uncle/niece



A. Simple types of consanguinity



$$\begin{aligned}
 I &\rightarrow III &= \left(\frac{1}{2}\right)^2 \\
 I &\rightarrow IV &= \left(\frac{1}{2}\right)^2 \\
 I &\rightarrow III \text{ and } IV &= \left(\frac{1}{2}\right)^4 \\
 III \text{ and } IV &\rightarrow V &= \left(\frac{1}{2}\right)^2 \\
 \text{IBD: } &\left[\left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4\right] \times \frac{1}{2} = \frac{1}{16}
 \end{aligned}$$

Probability of descent from a common ancestor

B. Identity by descent (IBD)

Twins

Twin, triplet, or quadruple pregnancies regularly occur in many species of animals. In humans, about 1 in 80 pregnancies results in twins. They can arise from a single fertilized egg (monozygotic twins, MZ) or from two different eggs (dizygotic twins, DZ). Monozygotic twins are genetically identical.

The study of genetic differences between monozygotic (identical) and dizygotic (nonidentical) twins seeks to determine the relative importance of genetic and environmental factors. While twin research has limited value in the search for individual genes, it may shed light on the possible genetic origin of complex traits, such as multifactorial diseases (see p. 154) and human behavior.

Comparisons of monozygotic twins who have been raised together with those who were raised apart are especially informative. They allow insight into the role of genetic factors as opposed to environmental influences. In a large study, the Minnesota Twin Study (Bouchard, 1990), monozygotic and dizygotic twins who had been separated since infancy were compared. A series of complex tests demonstrated that monozygotic twins raised apart, compared with those raised together, were very similar in spite of their different environmental influences. About 70% of the variance of mental ability was associated with genetic variation.

A. Types of twins

Monozygotic twins arise by separation at a very early stage of development. They always have a common placenta (monoplacental). They may lie in two amniotic cavities (diamniotic) or in a common amniotic cavity (monoamniotic). Dizygotic twins (DZ) always have their own amniotic cavity (diamniotic). They may each have their own placenta (diploplacental), or they may share a common placenta.

B. Pathological conditions in twins

Incomplete separation (so-called Siamese twins) or connections between the blood circulations lead to pathological conditions in twins. A relatively frequent form of incomplete separation is thoracopagus (1), in which the twins are joined to various extents at the thoracic region. The blood circulation of twins can be connected via the common placenta (formation of a

shunt) (2). This can lead to one twin receiving less blood or even bleeding to death. Especially severe malformations result from incompletely formed organs, e.g., absence of the heart in one twin (acardius) (3). If one of the twins dies very early in pregnancy, it may disappear by the time the other twin is born (vanishing twin).

C. Concordance or discordance

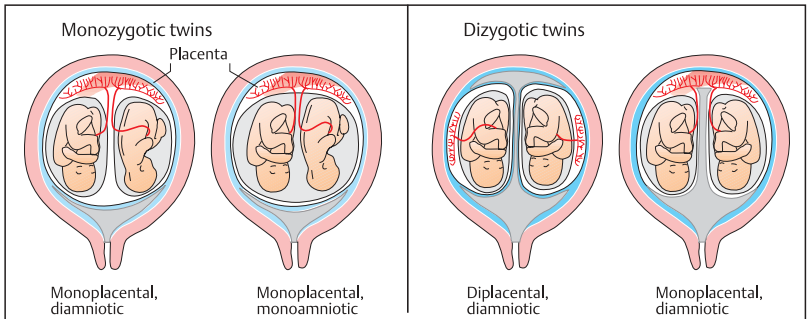
When twins show the same trait, they are said to be concordant; when they differ, discordant. Comparisons of the rate of concordance allow conclusions about the relative role of genetic factors in the etiology of complex traits such as some malformations, predisposition to diseases, height, blood pressure, etc. However, definitive conclusions about genetic causes can rarely be reached from concordance rates alone. (After Connor and Ferguson-Smith, 1991.)

D. Biochemical differences

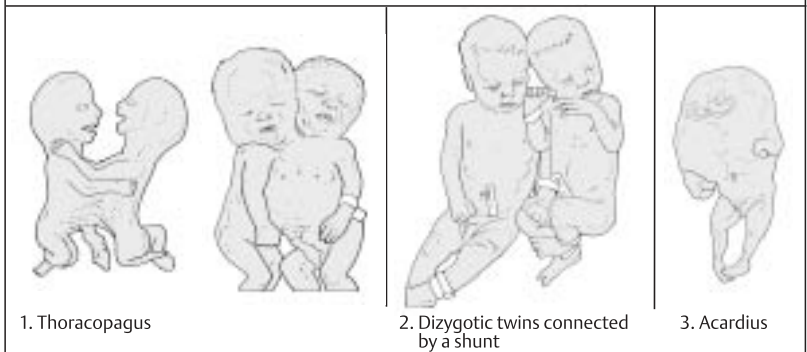
Dizygotic and monozygotic twins differ also biochemically. Due to genetic differences, many chemical substances used in therapy are metabolized or excreted at different rates, owing to different activities of corresponding enzymes (see p. 166 and 372). An example is the rate of excretion of phenylbutazole, as studied by E. S. Vesell. This substance is excreted at the same rate in identical twins, whereas the rates of excretion differ between dizygotic twins or siblings.

References

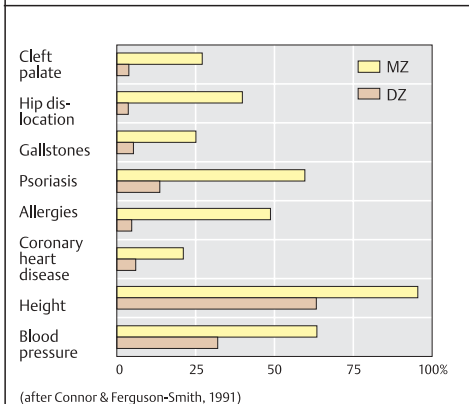
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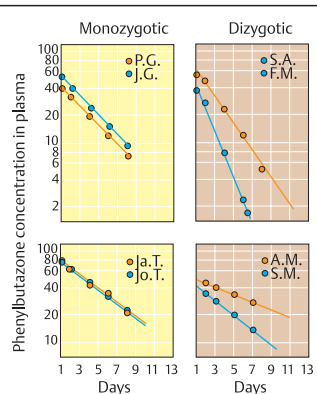
A. Types of twins



B. Pathological conditions in monozygotic twins



C. Concordance of some traits in monozygotic (MZ) and dizygotic (DZ) twins



D. Excretion rate of a drug (data of E. S. Vesell)

Polymorphism

In nature, organisms of the same species usually differ in some aspects of their appearance. The differences are genetically determined and are referred to as polymorphisms. Genetic polymorphism is defined as the occurrence in a population of two or more genetically determined alternative phenotypes due to different alleles, whereby the least frequent allele cannot be maintained by repeated mutation alone. A gene locus is defined as polymorphic if the rare allele(s) has (have) a frequency of at least 0.01 (1%), and as a result, heterozygotes for such alleles occur with a frequency of at least 2%. Polymorphism can be observed at the level of the whole individual (phenotype), in variant forms of proteins and blood group substances (biochemical polymorphism), in morphological features of chromosomes (chromosomal polymorphism), or at the level of DNA in differences of nucleotides (DNA polymorphism, p. 72). A polymorphism is considered as neutral if the presence or absence of a certain allele does not confer any advantage or disadvantage. A polymorphism may represent an advantage for the population, because it is likely to contain individuals who are better prepared for certain environmental conditions or changes thereof than if there were genetic uniformity.

A. Polymorphism of the phenotype

An impressive example of phenotypic polymorphism is the color pattern on the wing sheaths of the Asian ladybug (*Harmonia axyridis*) (1). In the area of distribution, extending from Siberia to Japan, multiple variants can be distinguished (Ayala, 1978). The different color combinations are due to different alleles of the same gene (1). In the California king snake (*Lampropeltis getulus californiae*) (2), color patterns differ to such an extent within the same species that they would seem to represent different species.

B. Polymorphism in relation to environmental conditions

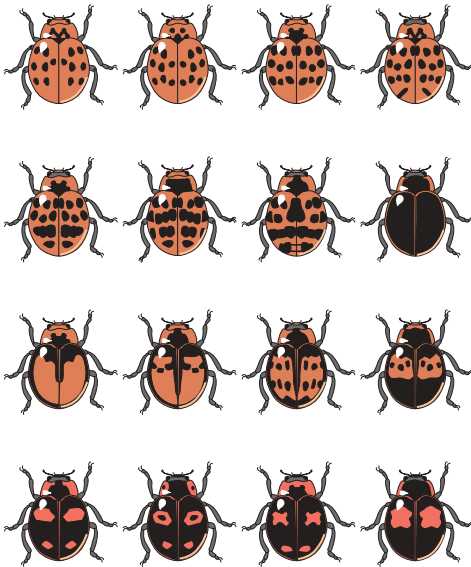
Most living organisms show geographical variation; that is, there are differences in the frequencies of alleles among populations of the same species in different areas. The differences may be gradual and reflect adaptation to environmental conditions. For example, the average height of the yarrow plant on the slopes of

the Sierra Nevada in California decreases as the altitude increases (1). By comparing the growth of plants whose seeds were obtained at different altitudes and sown in one garden, it can be shown that the average height of plants at the different altitudes is genetically determined. (Illustration after Campbell, 1990.)

One of the most impressive cases of natural selection for a polymorphic color pattern has been represented by the English peppered moth *Biston betularia*. In the midlands of England it was observed in two varieties, a light gray (*typica*), with spots of pigments, and a rare dark gray variety (*carbonaria*) (2). It was thought that the black variety was rare because it was easily seen by predators and therefore selected against. In the late 1880s, the light variety became rare and the black type prevailed. It was assumed that this was due to selection for *carbonaria*, which was camouflaged on the dark bark of trees after the light-colored lichens covering trees and rocks had been killed by the beginning industrial pollution. After pollution was reduced in the 1950s, the light type reappeared and again became common. Unfortunately, serious doubt has been cast on this beautiful story of natural selection by genetic adaptation. Majerus (1998) pointed out unconfirmed observations that make natural selection untenable in this case (see Nature, 5 November 1998, p. 35). What a pity. (Origin of illustrations: F. J. Ayala (A1), J. H. Tashjian, San Diego Zoo (A2), N.A. Campbell, Biology, 1990 (B1), Laßwitz and K. Jäkel, Okapia, Frankfurt (B2)).

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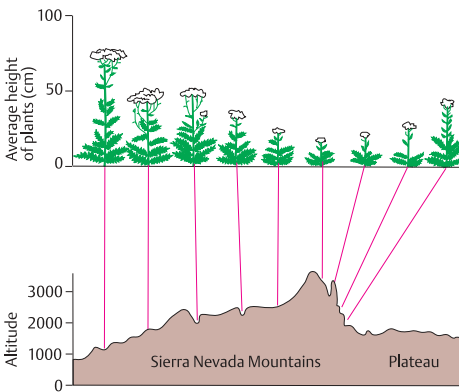


1. Asiatic beetle *Harmonia axyridis*

A. Polymorphism of the phenotype



2. California king snake
Lampropeltis getulus californiae



1. Yarrow plant in the Sierra Nevada Mountains



2. Moth *Biston betularia*,
light and dark type

B. Polymorphism in relation to environmental conditions

Biochemical Polymorphism

Biochemical polymorphism is detected by laboratory methods. It results from individual differences in the sequences of nucleotide bases of DNA. If a difference leads to a change in a codon, a different amino acid may be incorporated at the corresponding site of the protein. This can be demonstrated by analyzing the gene product.

Subtle differences in the many biochemical reactions between individuals was recognized early last century as the basis of genetic individuality by Archibald Garrod (Bearn, 1993).

A. Recognition of polymorphism by means of gel electrophoresis

Polymorphism of a protein (gene product) can be demonstrated by gel electrophoresis when one variant form differs from the others by the presence of an amino acid with a different electric charge. In this case, the allelic forms of the gene product can be distinguished owing to their different speeds of migration in an electrical field (electrophoresis). When a gene product consists of two subunits (dimeric protein), heterozygous individuals will show three bands. A polymorphism that does not lead to a change in electric charge cannot be identified in this manner.

B. Demonstration of polymorphism in gene products

Here, polymorphism is shown to be frequent in three typical gene products, the enzymes phosphoglucosutase, malate dehydrogenase, and acid phosphatase, in different species of *Drosophila*. Each of the diagram sections 1, 2, and 3 shows a starch gel electrophoresis of 12 fruit flies, each gel being specifically stained for the respective protein. With phosphoglucosutase (1) different migration speeds are observed in homozygous individuals as opposed to heterozygous (2, 4, 10). Malate dehydrogenase is a dimeric protein, so that heterozygous individuals (4, 5, 6, 8) show three bands. Acid phosphatase (3) shows a complex pattern because four alleles are involved. (Figure after Ayala and Kieger, 1984).

C. Frequency of polymorphism

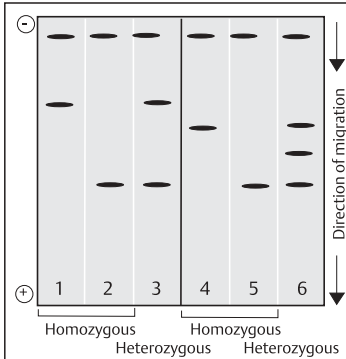
In a study of average heterozygosity in *Drosophila willistoni* (1), 17.7% of 180 gene loci were found to be heterozygous. The average heterozygosity is the proportion of heterozygous individuals in a population with reference to the number of analyzed loci. This is determined by summing the proportion of heterozygotes per total number of individuals per gene locus. The best insight into the considerable frequency of polymorphism is obtained from direct analysis of DNA. Hypervariable regions are found in many areas of the genome. Here, digestion with a restriction enzyme produces several different patterns due to individual differences in the size of the DNA fragments. In (2), the polymorphism at one locus near the H chain of the immunoglobulin J region in 16 individuals is shown (White et al., 1986). Hypervariable regions lead to an individual pattern for each person (DNA fingerprint).

D. Genetic diversity and evolution

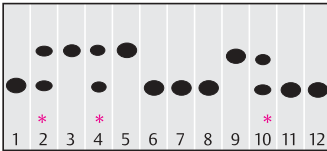
A population of genetically relatively homogeneous individuals is less likely to be able to adapt to changing environmental conditions than is a genetically diverse species. Two populations of *Drosophila serrata* kept for 25 generations in separate closed bottles with limited availability of space and nourishment were compared. The population with the greater genetic diversity adjusted better to the environmental conditions. (Figure after Ayala & Kieger, 1984).

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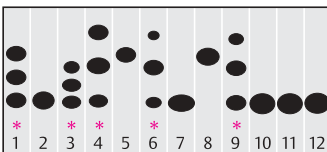
A. Recognition of polymorphism by gel electrophoresis



1. Phosphoglucomutase in *Drosophila pseudoobscura*

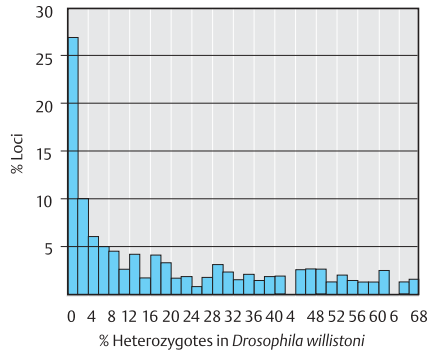


2. Malate dehydrogenase in *Drosophila equinoxialis*

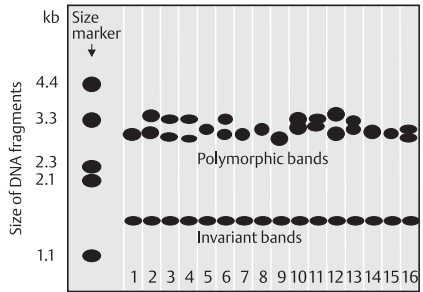


3. Acid phosphatase in *Drosophila equinoxialis* (* = Heterozygous fruit flies)

B. Demonstration of polymorphism in gene products

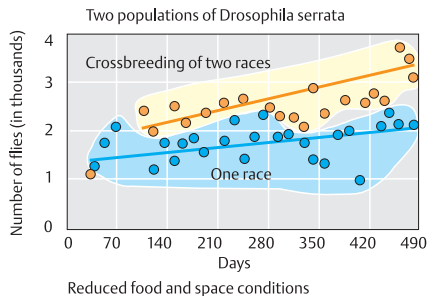


1. High-average heterozygosity (17.7 % of 180 loci)



2. DNA polymorphism in man (16 individuals)

C. Frequency of polymorphism



D. Genetic diversity and evolution

Geographical Distribution of Genes

Within a species, alleles at a given gene locus may occur with different frequencies in different geographical regions. This can be due either to random differences in frequencies (genetic drift) or to selection. Selection means that one allele has conferred an advantage over another (selective advantage), with the result that the individual has a slightly higher probability of survival in that particular environment.

A. Different frequencies of hereditary disorders, e.g., in Finland

Genetically determined diseases in humans (hereditary diseases) occur with different frequencies in different regions of the world. For example, the distribution of the grandparents of persons affected with three autosomal recessive inherited diseases in Finland shows an increased regional clustering because each mutation has arisen in a different region of Finland: (1) congenital flat cornea, in the western part of the country; (2) the Finnish type of congenital nephrosis (a severe renal disorder), in the southern part; and (3) diastrophic skeletal dysplasia, in other southern regions. There is no known basis for a selective advantage of these mutant alleles in the heterozygotes to explain their frequencies. The different distribution merely reflects the place and relative point in time of the mutation. The mutations must have arisen after Finland was settled, since they are unusually frequent there but rare elsewhere.

Similar examples are found in many other regions of the world and in other populations.

B. Distribution of malaria and frequency of heterozygotes due to selective advantage

The distribution of malaria in the Mediterranean region, West Africa, Sub-Saharan Africa, and East Africa including the Nile Valley and Asia (1) is similar to the frequency of heterozygotes for three forms of genetic disease: sickle cell anemia, thalassemia (see p.340 ff.), and glucose-6-phosphate dehydrogenase deficiency (a red-cell disorder). Heterozygotes for these three genetically determined diseases have a selective advantage here. They are rela-

tively protected because conditions in the blood of heterozygotes are less favorable for the malaria parasite than in the blood of normal homozygotes (see p.340).

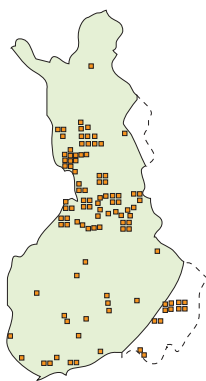
Glucose-6-phosphate dehydrogenase deficiency (2) is an X-chromosomal disorder that leads to severe illness in affected males (hemizygotes). In the same way, homozygotes for sickle cell anemia (3) and different forms of a disorder of hemoglobin formation (thalassemia, 4) are very severely affected, whereas heterozygotes are not or are only mildly affected. This has led to a striking increase of heterozygotes for the mutant allele in areas infested with malaria.

Protection for the heterozygotes occurs at the cost of affected homozygotes, but has a positive effect for the population as a whole, since the total frequency of affected individuals is reduced. Nevertheless, the price of this advantage is considerable. More than 400 million people become ill with malaria each year in Africa, Asia, and South America. About 1 to 3 million die. It is hoped that sequencing the malaria-causing parasite, *plasmodium falciparum*, and other approaches will invigorate the fight against malaria (Malaria New Focus, Science 20 October 2000, p.428–441).

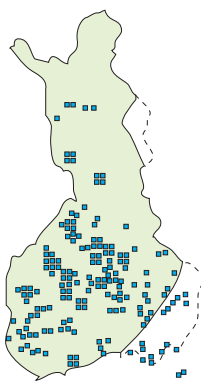
The sickle cell mutant has arisen independently at least four times in different regions and has become established there due to selective advantage. The situation is similar for thalassemia (see p.344).

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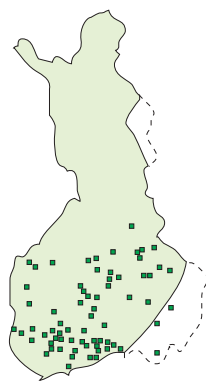
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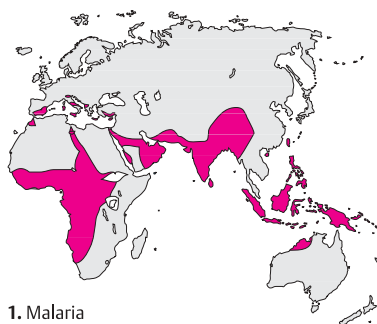
1. Congenital corneal dystrophy



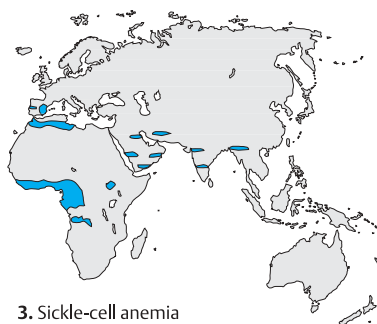
2. Congenital nephrosis



3. Diastrophic skeletal dysplasia

A. Different frequency of genetic diseases, e.g., in Finland

1. Malaria



3. Sickle-cell anemia



2. Glucose-6-phosphate dehydrogenase deficiency



4. Thalassemia (different forms)

B. Distribution of malaria and frequency of heterozygotes by selective advantage

Chromosomes

Nucleosomes

The total length of DNA in the haploid chromosome complement in nondividing cells of mammals corresponds to about 1 meter. Since the total length of the human haploid chromosome complement in metaphase cells corresponds to about 115 μm , DNA in metaphase chromosomes must be shortened 10 000-fold by very efficient packing. This occurs in discrete units, the nucleosomes, consisting of DNA and histones.

A. DNA and histones

Histone proteins (histones) are proteins with a high proportion of positively charged amino acids (lysine and arginine), which enable them to bind firmly to the negatively charged DNA double helix. There are five types of histone molecules: H1, H2A, H2B, H3, and H4. Except for H1, they occur in exactly equal numbers. Histones are evolutionarily highly conserved. There is an especially close spatial relationship between DNA and histones in the A+T-rich regions of the minor grooves. Here, the DNA double helix is most easily bent. (It is bent more than shown here).

B. Nucleosome

The nucleosome is the fundamental subunit of DNA and histones in interphase chromosomes, or chromatin (see next plate). A nucleosome consists of about 146 bp of DNA wound twice around an octamer of histones, two copies each of H2A, H2B, H3, and H4, the core histones. The histone octamer forms a cylinder of 11 nm diameter and 6 nm height (the figure in B is not to scale and is highly schematic; for details see Lewin, 2000).

C. Three-dimensional structure of a nucleosome

The ribbon diagram of the nucleosome based on the X-ray structure at high resolution shows one DNA strand in green, the other in brown. The histone are shown in different colors in a view from above (figure adapted from Luger et al., with kind permission by T. J. Richmond).

D. Chromatin structures

Chromatin occurs in condensed (tightly folded), less condensed (partially folded), and extended

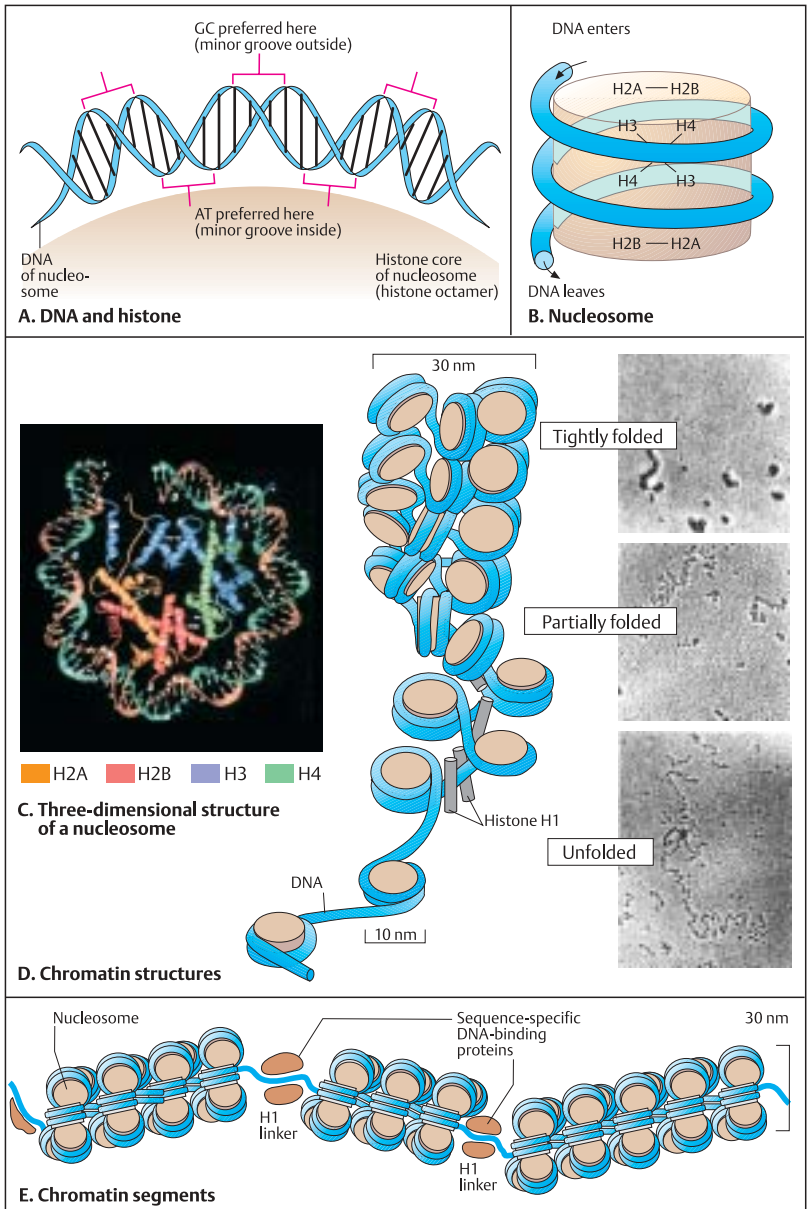
unfolded form. When extracted from cell nuclei in isotonic buffers, most chromatin appears as fibers of about 30 nm in diameter. In less condensed chromatin fibers, it becomes apparent that DNA between nucleosomes is bound to H1 histones. H1 histones vary among tissues and represent about half of the amount of individual core histones. The corresponding electron-microscopic photographs obtained by different techniques show the condensed (folded) chromatin as compact 300–500 Å structures (top), a 250 Å fiber when partially folded, and as a "beads on a string" 100 Å chromatin fiber (bottom). Condensed chromatin is referred to as heterochromatin, extended chromatin as euchromatin (see following pages). (Diagram adapted from Alberts et al. 1994; the electron-micrographs are from Thoma and Koller, 1979).

E. Chromatin segments

The chromatin of the nucleus consists of segments of nucleosomes condensed to different degrees. DNA wound around a nucleosome is inactive and unreactive. In the DNA segments lying between series of nucleosomes, sequence-specific DNA-binding proteins are found. They regulate the expression (activity) of genes (see pp.204 ff). (Figures after Alberts et al. 1994).

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DNA in Chromosomes

In eukaryotic organisms the DNA is contained in chromosomes. Chromosomes consist of DNA and its closely associated proteins, which are histones and nonhistone proteins in about equal amounts. The chromatin of the cell nucleus is composed of chromosomes in interphase. Chromosomes are visible as individual structures only during mitosis. The haploid genome in humans contains 3×10^9 base pairs in 22 autosomes and two sex chromosomes (X and Y).

A. DNA in metaphase chromosomes

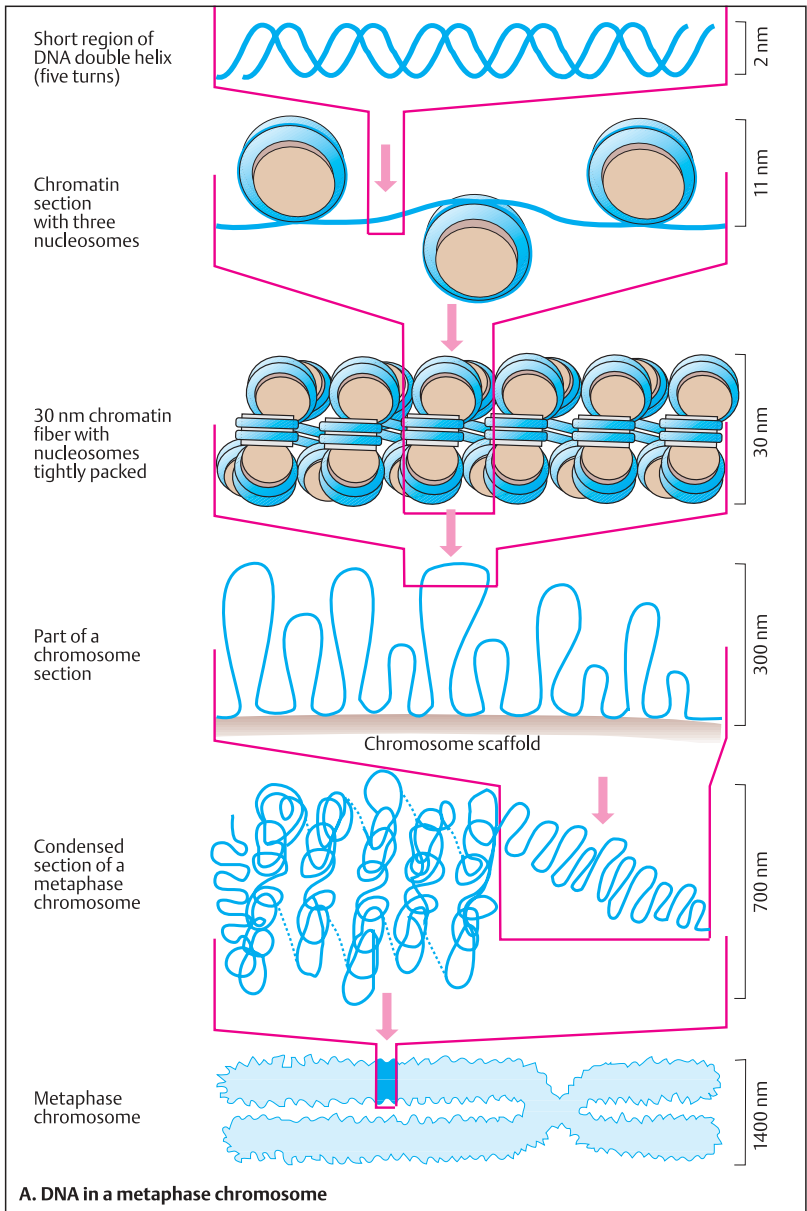
A gene is a nucleotide sequence that serves as the functional unit for the formation of a complementary RNA molecule (transcription unit). A chromosome consists of a single, very long DNA molecule that contains a series of genes. It is folded and coiled in a complex manner. The spatial arrangement is secured by the nucleosomes.

In eukaryotic cells, the DNA is closely associated with histones to form characteristic structures, the nucleosomes. In addition, there are hundreds of sequence-specific DNA-binding proteins that recognize short DNA segments. Their binding is an important means of regulating gene activity (gene expression, p. 204ff.). Chromosomal DNA is folded and packed in an efficient manner. Schematically, several levels of packing of DNA in a metaphase chromosome can be differentiated. The figure shows (at the top) a segment of DNA double helix from part of a chromatin segment. The chromatin segment consists of several nucleosomes connected by so-called linker stretches of DNA. The folding of a chromatin segment produces a chromatin fiber of 30 nm diameter, visible on electron microscopy. It consists of a series of tightly packed nucleosomes. These in turn form a part of a chromosome segment of about 300 nm diameter in the extended stage. A further packing level is represented by a thickened segment of a metaphase chromosome. This is called a condensed chromosomal segment. This segment corresponds to only a small part of one chromatid of a metaphase chromosome. As described on p. 182, each metaphase chromosome consists of two chromatids (sister chromatids), one of these chromatids being the result of DNA replication in the S phase. No transcription oc-

curs in the chromatin of mitotic chromosomes. RNA synthesis is halted with the onset of chromosome condensation. (Figure after Alberts et al., 1989).

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Polytene Chromosomes

In certain cells of some insects, chromatin structures are visible under the light microscope as individual looplike domains. These cells and the chromosomes in them are greatly enlarged by prior DNA synthesis without cell division. That is, they contain about a thousand times more DNA than usual. The multiple copies of homologous chromosomes remain side by side in these cells and form a gigantic polytene chromosome. If the multiple copies of a polytene chromosome did not occur side by side, but rather as discrete chromosomes, the result would be referred to as polyploidy. Since there is a direct transition from polytene to polyploid chromosomes, the polytene chromosomes probably correspond in structure to normal chromosomes. Polytene chromosomes are readily visible under the light microscope because they are so large and lie precisely next to each other.

A. Polytene chromosomes in the salivary glands of *Drosophila* larvae

Under the light microscope, polytene chromosomes show readily recognized patterns of alternating dark and light bands. The light bands are called interbands. A polytene chromosome results from ten cycles of replication without division into daughter chromosomes. Thus, there are about 1024 (2^{10}) identical chromatid strands, which lie strictly side by side. About 80% of the DNA in polytene chromosomes is located in bands, and about 15% in interbands. The chromatin in the darkly stained band is condensed to a much greater degree than the chromatin in the interbands. This can be attributed to an especially high degree of folding. Depending on the size, an individual band contains between 3000 and 300 000 nucleotide base pairs. Altogether, the *Drosophila* genome contains about 5000 bands and the same number of interbands. They have been numbered to produce a polytene chromosome map. A microscopic detail of a polytene chromosome from a *Drosophila* salivary gland shows the characteristic band pattern. The dark bands are the result of chromatin condensation in the large interphase polytene chromosomes. These bands must be differentiated from those visible only in metaphase after special staining techniques in the banding patterns of the karyotypes of

eukaryotic organisms (see p. 184). Intensely stained chromosomal segments correspond to a high degree of packing and are genetically inactive (heterochromatin); less tightly packed segments stain less distinctly and correspond to segments with genetic activity (euchromatin).

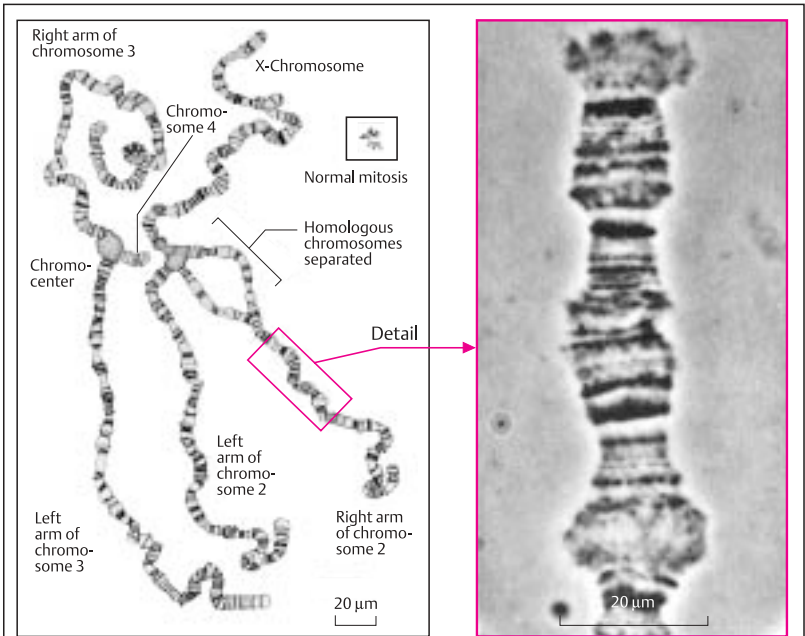
B. Functional stages in polytene chromosomes

Polytene chromosomes form structures that correlate with the functional state: During the larval development of *Drosophila*, a series of expansions (puffs) appear in temporal stages in the polytene chromosomes. Chromosome puffs are decondensed, expanded segments that represent active chromosomal regions, i. e., regions that are being transcribed. The location and duration of the puffs reflect different stages of larval development (1). The incorporation of radioactively labeled RNA has been used to demonstrate that RNA synthesis, a sign of gene activity (transcription), occurs in these regions (2).

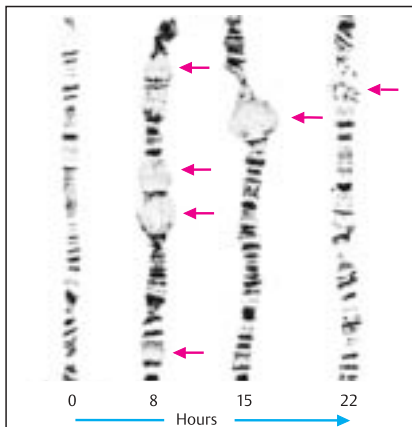
(Figures after Alberts et al., 1994 and Watson et al., 1987).

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A. Polytene chromosomes in salivary glands of *Drosophila* larvae



1. Formation of puffs (arrows)



2. Evidence of gene activity

B. Functional stages in polytene chromosomes

DNA in Lampbrush Chromosomes

Usually, the chromatin in interphase nuclei is so densely packed that single chromatin threads are not directly visible. An exception is in the interphase nuclei of certain cells, especially immature amphibian oocytes with paired meiotic chromosomes. These chromosomes show RNA synthesis and form unusual chromatin loops. They are covered with newly transcribed RNA and are packed in dense RNA–protein complexes. These are visible under the light microscope as the so-called lampbrush chromosomes.

A. Chromosome structure in amphibian oocytes (lampbrush chromosomes)

The paired chromosomes of oocytes in meiosis consist of numerous chromatin loops arranged along an axis (1). Chiasma formation is visible at various locations.

Each segment of a lampbrush chromosome consists of a series of chromatin loops, originating from an axis (2). There, a condensed structure, the chromomere, is visible. Any given chromatin loop always contains the same DNA sequence and is formed in the same manner as the oocyte grows. Each loop corresponds to a particular unit of unfolded chromatin, which has unraveled and is transcriptionally active. Each cell contains four copies of a loop, two per chromosome (four-strand structure).

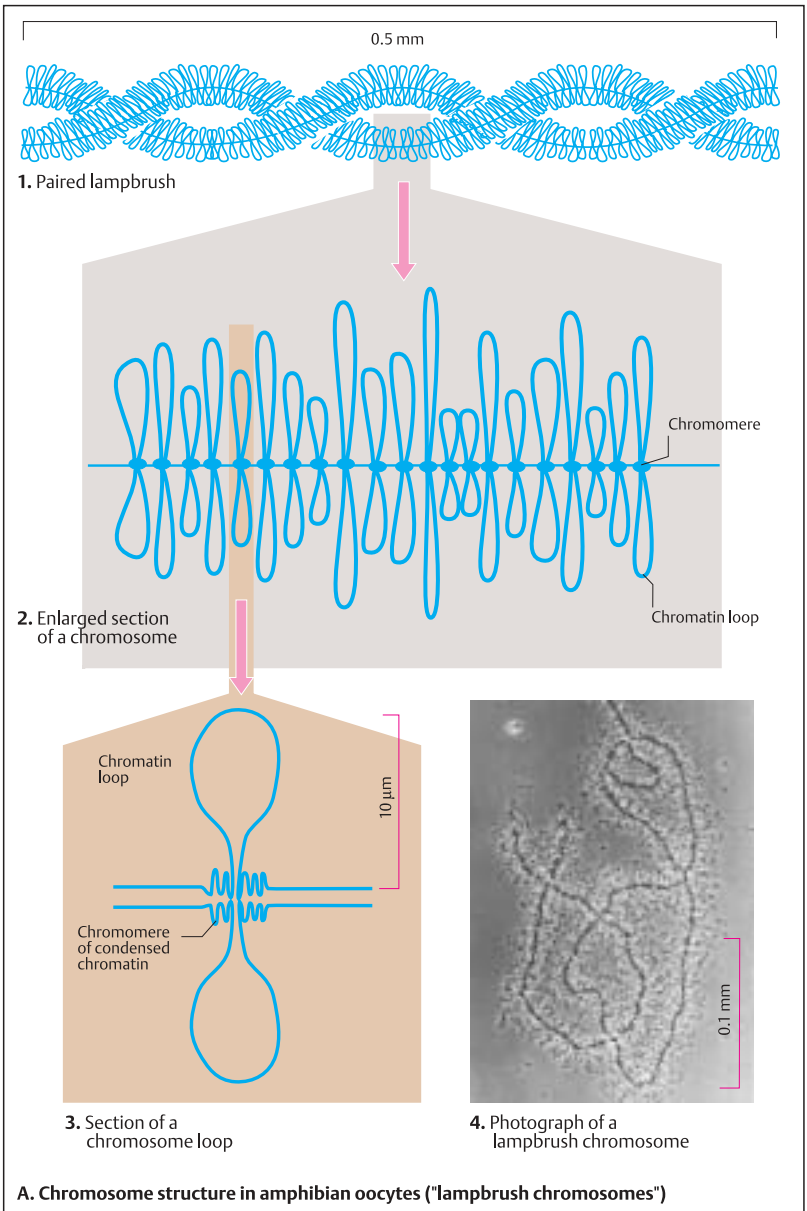
Transcription occurs either along the whole loop or at parts of a loop. The chromatin loops of a chromosome are paired, mirror-image structures (3). Each corresponds to the loop of a sister chromatid. The chromomere at the base of the loops consists of dense chromatin of the two sister chromatids, while the chromatin loops themselves are not condensed.

Lampbrush chromosomes in amphibian oocytes are unusually large compared with mitotic chromosomes (about 400 μm long, as opposed to at the most 10 μm) as shown in the micrograph (4, phase contrast exposure, Gall 1956). At the beginning of meiosis, when DNA replication is complete, the homologous pairs lie immediately next to each other and form characteristic structures composed of four chromatids (p. 116).

Lampbrush chromosomes are distinguished by an especially high rate of RNA transcription. Most of the RNA transcripts are longer than in other chromosomes. (Figures redrawn from Alberts et al., 1994).

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Correlation of Structure and Function in Chromosomes

In a chromosome, regions with special structures and functions can be differentiated. The centromere is the point of attachment of the microtubuli of the mitotic spindle (kinetochore). The telomeres at the ends contain no genes and have a special structure.

A. Heterochromatin and euchromatin

In 1928 Emil Heitz observed that certain parts of the chromosomes of a moss (*Pellia epiphylla*) remain thickened and deeply stained during interphase, as chromosomes otherwise do only during mitosis. He named these structures heterochromatin, as opposed to euchromatin, which becomes invisible during late telophase and subsequent interphase. Functionally, heterochromatin is defined as a region in which few or no active genes lie and in which repetitive DNA sequences occur. When active genes become located close to the heterochromatin, they usually become inactivated (position effect-variegation). (Figure from Heitz, 1928).

B. Characteristic regions of a chromosome

The centromere and telomeres contain repetitive DNA sequences. They are evolutionarily conserved because they are important for chromosome stability. The segments located between the telomeres and centromere consist of trypsin-sensitive light and trypsin-resistant dark G-bands. The light G-band areas of DNA form loops in which the protein-coding genes lie. The DNA loops are bound to a protein matrix at special attachment sites.

C. Model of a chromosome segment in interphase

A three-dimensional model of a chromosome segment shows that the constitutive heterochromatin (C-band) in the centromere region is very tightly wound. In the light G-bands, the euchromatin is relatively loosely packed, and in the dark G-bands, somewhat more tightly packed. (With kind permission of the author, from Manuelidis, 1990, copyright 1990 by the AAAS).

D. Constitutive heterochromatin (C-bands) in the centromeric region

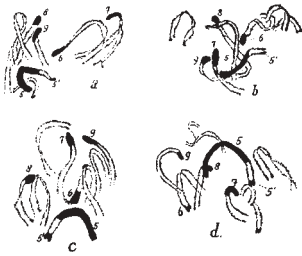
The constitutive heterochromatin in the centromeric region can be specifically stained (C bands). The distal half of the long arm of the Y chromosome is also C-band positive. The centromeric heterochromatin in chromosomes 1, 9, and 16 and in the long arm of the Y chromosome in humans is polymorphic. The lengths of the heterochromatic segments in one or more of these regions may vary among different individuals. (From Verma and Babu, 1989).

E. Functional attributes of the euchromatin regions

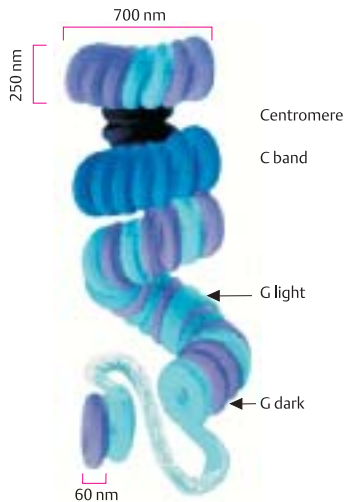
The light and dark G-bands differ in functional respects. An average G-band contains around 1.5 megabases (Mb) of DNA.

References

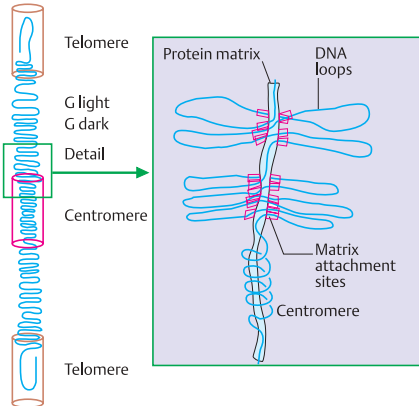
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A. Heterochromatin and euchromatin



C. Model of a chromosomal segment in interphase



B. Special parts of a chromosome



D. Constitutive heterochromatin (C bands) at the centromeres

Light G bands

GC-rich
Fluorescence with G-specific fluorochromes, e.g., olivomycin
Early replicating
Gene-rich
Alu repeats
SINE repeats
(short interspersed repetitive DNA sequences)
Z-DNA conformation possible

Dark G bands

AT-rich
Fluorescence with AT-specific fluorochromes, e.g., quinacrine
Late replicating
Gene-poor
LINE repeats
(long interspersed repetitive sequences)
HMG-1 nonhistone proteins bound to AT-rich areas
Minisatellites

E. Functional properties in euchromatin

Special Structure at the Ends of a Chromosome: the Telomere

Unlike the circular chromosomes of bacteria, bacteriophages, plasmids, and mitochondrial DNA, the chromosomes of eukaryotes are linear. Each end is "sealed" by a specialized region, the telomere. Telomeres stabilize chromosomes at both ends.

A. Replication problem at the ends of linear DNA

Since DNA is synthesized in the 5' to 3' direction only, the two templates of the parent molecule differ with respect to the continuity of synthesis. On the 5' to 3' template strand, synthesis occurs in the reverse direction relative to the fork movement (lagging strand synthesis). There, DNA is synthesized in short fragments about 1000–2000 nucleotides long in bacteria and about 200 nucleotides in eukaryotes (Okazaki fragments, see DNA replication, p. 42). However, 8–12 bases at the end of the lagging strand template cannot be synthesized by DNA polymerase because the primer it requires cannot be attached beyond the end of the template strand. Hence, at each round of replication before cell division, these 8–12 nucleotides will be lost at the chromosome ends. Some organisms compensate for this loss by adding telomeric repeats to the ends of the chromosome during the replication cycle.

B. G-rich repetitive sequences at the telomeric region

DNA at the telomeres consists of G-rich tandem sequences (5'-TTAGGG-3' in vertebrates, 5'-TGTGGG-3' in yeast, 5'-TTGGGG-3' in protozoa). The G-strand overhangs are important for telomeric protection by formation of a duplex loop (see C).

C. Telomerase activity and stabilization by a loop

Two features characterize the telomere: telomerase activity to compensate for replication-related loss of nucleotides at the chromosome ends and telomeric DNA loop formation to stabilize the chromosome ends. Telomerase is a modified reverse transcriptase consisting of protein and about 450 nucleotides of RNA. Near the RNA 5' end are sequences complementary to

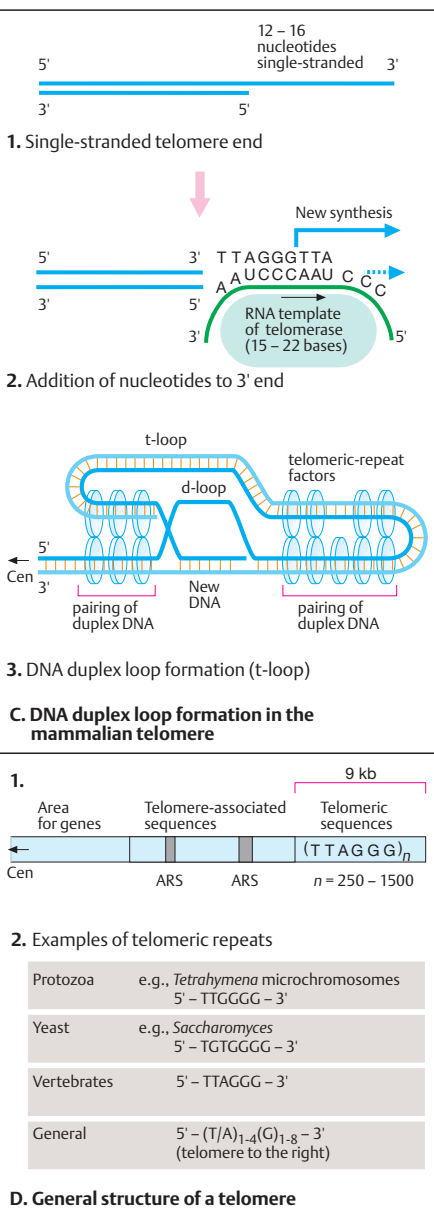
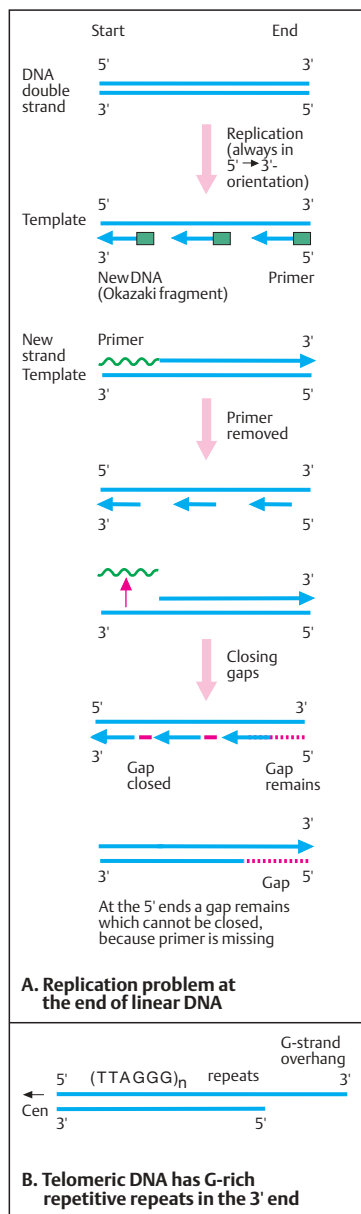
telomeric DNA repeat sequences. A short nucleotide sequence of this RNA pairs with terminal DNA sequences. The adjacent RNA nucleotides provide the template for adding nucleotides to the 3' end of the chromosome. After telomerase has extended the 3' (G-rich) strand, a new Okazaki fragment can be synthesized at the 5' strand by DNA polymerase. Griffith et al. (1999) have shown that telomeric duplex DNA forms a loop (t-loop), thus avoiding the "sticky end" problem. The loop formation is mediated by the two related proteins TRF1 (telomeric repeat-binding factor) and TRF2, which bind to mammalian telomere repeats, and the loop is anchored by the insertion of the G-strand overhang (see B) into a proximal segment of duplex telomeric DNA.

D. General structure of a telomere

In the terminal 6–10 kb of a chromosome, telomeric sequences and telomere-associated sequences can be differentiated (1). The telomere-associated sequences contain autonomously replicating sequences (ARS). The telomere sequences consist of about 250 to 1500 G-rich repeats (≈ 9 kb). They are highly conserved among different species (2). Telomerase activity is essential for survival in protozoans and yeast. In vertebrates it occurs mainly in germ cells, and no telomerase activity is found in somatic tissues. The cell division-dependent decrease of telomere length is viewed as being related to aging and death of cells because, ultimately, functional DNA will be lost. Unlike normal cells, many tumors have telomerase activity.

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Metaphase Chromosomes

Chromosomes are visible as separate structures only during mitosis. In interphase, the chromosomes in chromatin cannot be individually differentiated. In metaphase chromosomes, DNA is packed about 10 000 times more densely than in interphase. Electron-microscopic studies of metaphase chromosomes have yielded some insight into chromosomal structure.

A. A histone-free chromosome under the electron microscope

When certain proteins, especially histones, are removed from chromosomes, the chromosomal skeleton becomes visible under the electron microscope (1). Such a structure is surrounded by numerous darkly stained threads. A higher magnification (2) shows that this is a single continuous thread. It corresponds to the DNA double helix. (Photographs from Paulson and Laemmli, 1977.)

B. The microscopic appearance of metaphase chromosomes of man

With an approximately 1000-fold magnification, the metaphase chromosomes of man and other vertebrates can readily be recognized under the light microscope as individual rodlike structures. A metaphase is shown here at about 2800-fold magnification. The chromosomes differ from each other in length, in the size and arrangement of their transverse light and dark bands (banding pattern), and in the point of attachment of the spindle (centromere), which is recognizable as a constriction. In prometaphase, the chromosomes are longer than in metaphase and show more bands. Thus, for certain purposes chromosomes are also studied in prometaphase.

C. Types of metaphase chromosomes

Depending on the location of its centromere (point of attachment of the spindle during mitosis), a chromosome can be distinguished as submetacentric, metacentric, acrocentric, or telocentric. The centromere divides a submetacentric chromosome into a short arm (p arm) and a long arm (q). In metacentric chromosomes, the short and long arms are about the same length. Acrocentric chromosomes show a dense appendage called a satellite (not to be confused with satellite DNA) at the end of the

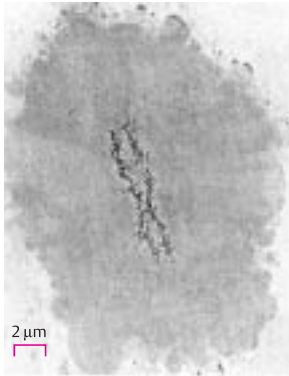
short arm. Satellite size differs for each acrocentric chromosome of an individual (chromosomal polymorphism). Telocentric chromosomes have neither a short arm nor a satellite. None of the human chromosomes are telocentric, whereas all chromosomes are telocentric in the house mouse, *Mus musculus*. However, it is debatable whether telocentric chromosomes actually exist as defined.

D. Simple structural aberrations

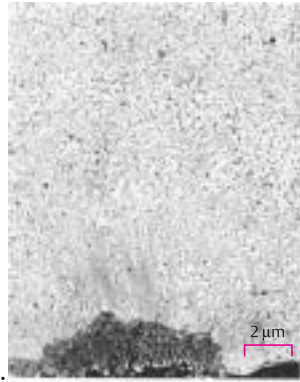
A functionally relevant deviation from the normal structure is called a structural aberration. This is to be differentiated from chromosomal polymorphism. Loss (deletion) or doubling (duplication) of a particular segment may occur. A deletion may occur at the end of a chromosome (terminal deletion) or within a chromosomal segment (interstitial deletion). Prerequisite for a terminal deletion is one break; for an interstitial deletion, two breaks. A segment that has been doubled is called a duplication. In metaphase chromosomes, an aberration is seen in both chromatids because as a rule it has occurred before the S phase. Duplications and deletions represent opposite, and in some respects complementary, aberrations of chromosomal structure (p. 404).

References

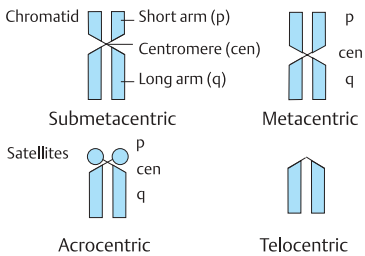
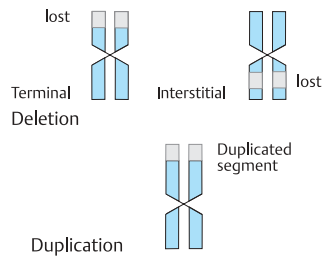
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1.



2.

A. Histone-free chromosome in the electron microscope**B. Human metaphase chromosomes****C. Types of metaphase chromosomes****D. Simple types of structural aberrations**

Karyotype

Karyotype refers to the arrangement of chromosomes in homologous pairs. They are arranged and numbered according to a convention. The basis for the arrangement is size of a chromosome, position of the centromere, and the chromosome-specific banding pattern. The karyotype is characteristic for each species. However, the term karyotype can also be applied to an individual or to a single cell.

A. The karyotype of man

Man (*Homo sapiens*) has 22 pairs of chromosomes (autosomes) and in addition either two X chromosomes, in females, or an X and a Y chromosome, in males (karyotype resp. 46,XX or 46,XY). In front of the comma, the karyotype formula gives the total number of chromosomes present, and after the comma, the composition of the sex chromosomes. The 22 pairs of autosomes in man are divided into seven groups (A–G) (see p. 186–190).

B. Karyotype of the mouse (*Mus musculus*)

The standard karyotype of the mouse consists of 19 chromosome pairs in addition to the X and Y chromosomes. All chromosomes except the X and Y are telocentric and of similar size (1). However, they differ in their banding patterns, characteristic for each chromosome pair, and therefore are individually distinct. Certain strains of mice may show variants of the karyotype (2). These variants arise from fusion of certain of the chromosomes. In the example shown here, only chromosome pairs 1, 15, 19, and X correspond with those of the standard karyotype, while the others consist of fused chromosomes, e. g., chromosomes 4 and 2, chromosomes 8 and 3, etc. Structural rearrangements of the karyotype occurred with the separation of different species in evolution. (Figure from Traut, 1991; photographs by H. Winking, Lübeck.)

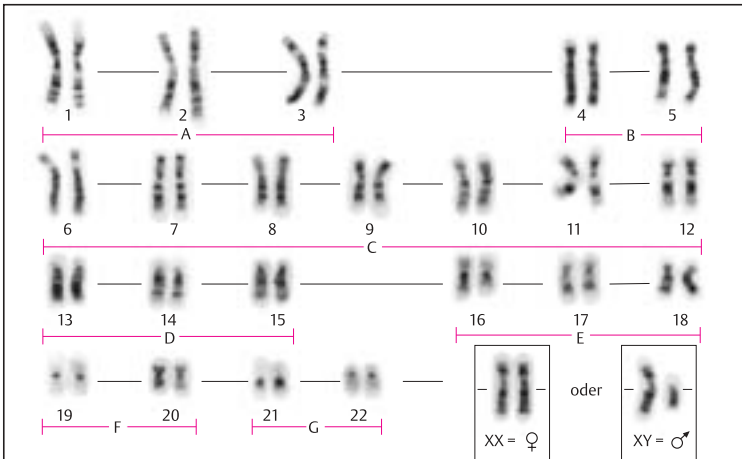
C. Flow cytometry karyotype in man

Because of their different lengths, metaphase chromosomes can also be presented in a flow-cytometry-based karyotype. With this method, individual chromosomes, stained and passed by a laser light source, give signals corresponding to their sizes. Although there are overlaps, e. g.,

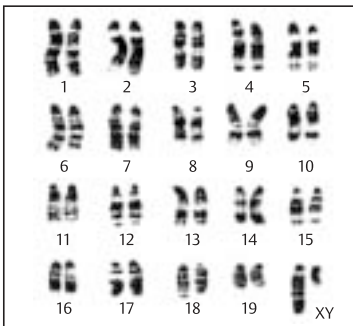
between the similarly sized human chromosomes 9–12, or with chromosomes 1 and 2, a distribution pattern of light impulses based on individual chromosome sizes is obtained. The size of the X chromosome lies between those of chromosomes 8 and 7; the size of the Y chromosome as a rule corresponds to that of a chromosome 22, although Y chromosome sizes may differ considerably. Because of the technical expenditure required and the unsharp resolution, flow cytometry is not frequently used for a practical diagnosis. However, it is of advantage in certain studies. (Figure from Connor and Ferguson-Smith, 1991.)

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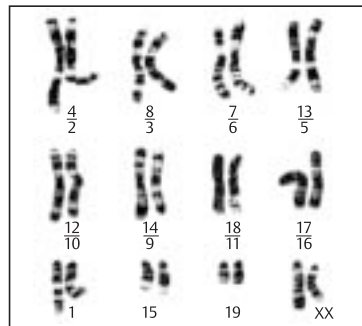
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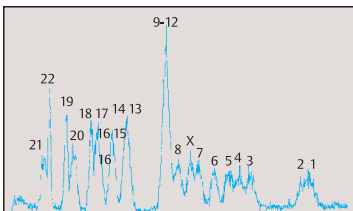
A. Karyotype of man



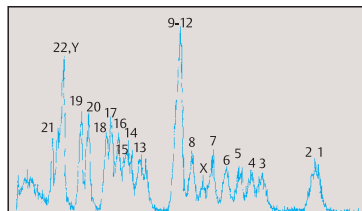
1. Standard



2. Variants in a population with fused chromosomes

B. Karyotype of the mouse (*Mus musculus*)

1. Normal female



2. Normal male

C. Flow cytometry karyotype

The G- and R-Banding Patterns of the Human Metaphase Chromosomes

With certain preparation and staining procedures, it is possible to distinguish light and dark bands in metaphase and prometaphase chromosomes under the light microscope. Each chromosome pair and each chromosome segment of sufficient length has a recognizable band pattern, specific for the respective chromosome. The basic types of bands are G-bands (Giemsa stain-induced) and R-bands (reverse bands). A procedure used for fluorescence microscopy produces quinacrine-induced bands (Q-bands). Their pattern corresponds to that of G-bands. Pages 187 and 189 show photographs of human metaphase chromosomes with G-bands (left) and R-bands (right). Between them, the corresponding G-band and R-band patterns are presented diagrammatically. Starting at the centromere, each chromosome is divided into defined regions and bands. Only the G-band-system numbering is shown. About 550 individually distinguishable bands of the haploid chromosome complement can be recognized in the illustration (550-band stage).

A. The metaphase chromosomes of man (pairs 1–22, X and Y)

The 22 pairs of autosomes (1–22) are divided into seven groups. Chromosome pairs 1–3 form

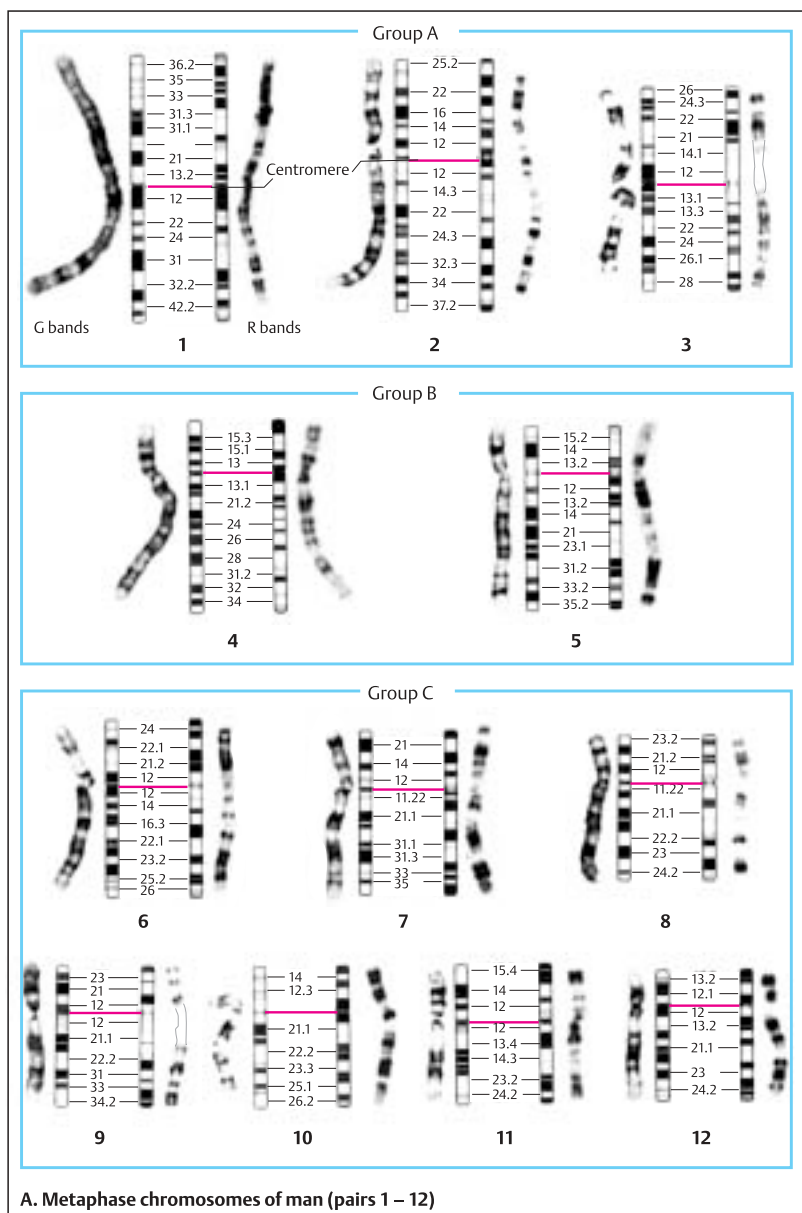
group A (chromosomes 1 and 3 are metacentric; chromosome 2 is submetacentric); chromosomes 4 and 5 form group B; 6–12 group C; and so on. The acrocentric chromosomes in humans belong to groups D and G. The chromosomes of group F are metacentric. The possibility of identifying every chromosome region and many bands makes it possible to establish breakpoints and to state the localization of genes according to region and band number on each chromosome. In good preparations subdivisions can be defined by decimals. For instance, band 31 includes bands 31.1, 31.2, and 31.3. (Figures correspond to Figure 5 from Harnden and Klinger, 1985, in *Birth Defects: Original Article Series*, vol. 21, no. 1, March of Dimes Birth Defects Foundation, New York, 1985. The G-band photographs are from Francke, 1981; and the R-band photographs are from Camargo and Cervenka, 1982.)

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The main types of chromosome bands

Banding methods	Type	Principal use
Trypsin-induced Giemsa stain	G	Differentiates light and dark bands
AT-specific fluorochrome (quinacrine, Hoechst 3325B)	Q	Light fluorescence in the region of dark G-bands, some centromere regions, distal long arm of the Y chromosome
Reverse bands	R	Opposite of G
Centromere stain	C	Centromere region darkly stained
Bromodeoxyuridine (BrdU) for two cell cycles	SCE	Differential staining of sister chromatids (sister chromatid exchanges)
Distamycin A–DAPI	DA/DAPI	Light fluorescence in the short arm of chromosome 15, centromere regions of 1, 9, and 16; distal long arm of Y
Silver nitrate stain	NOR	Short arms of all acrocentric chromosomes
Giemsa 11	G11	Centromere of chromosome II

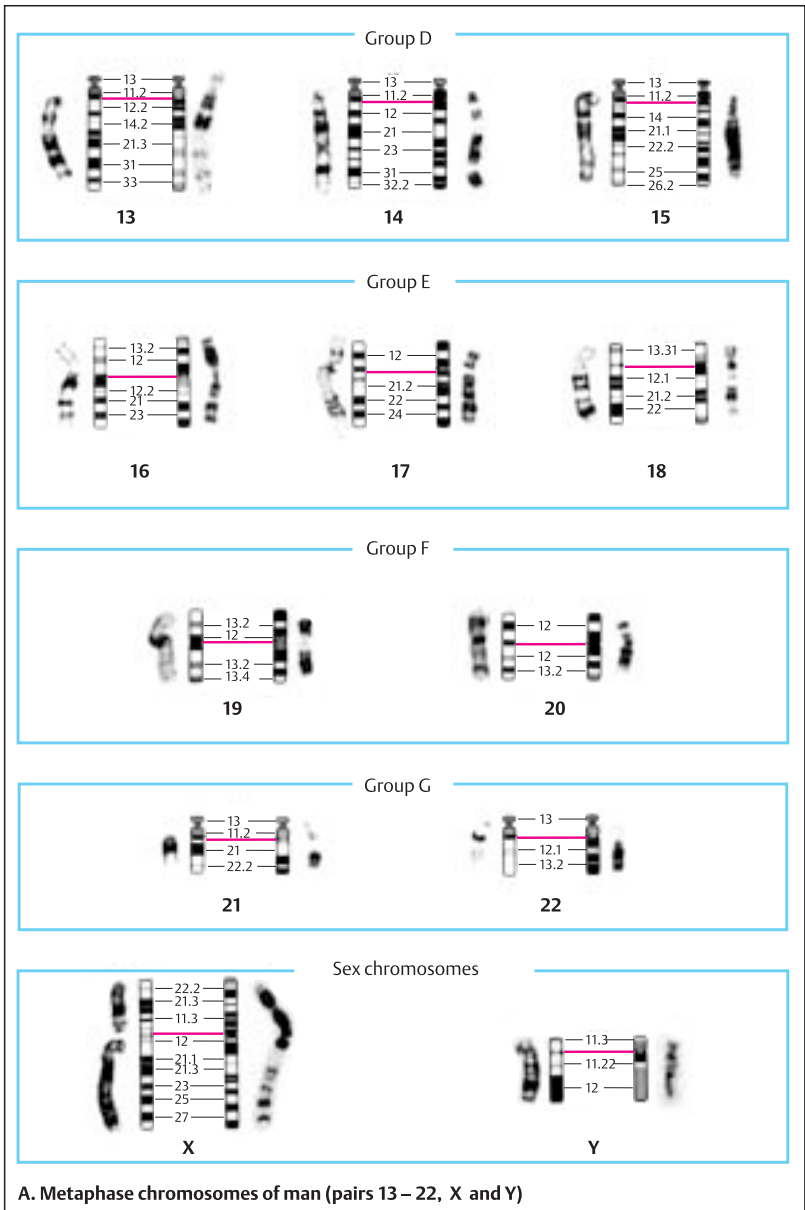


Designation of Chromosomal Aberrations

The following list of chromosome abbreviations includes the main types of chromosomal aberrations in man, abbreviated according to the In-

ternational System for Human Cytogenetic Nomenclature (ISCN 1995).

46,XX	Normal female karyotype with 46 chromosomes (two X chromosomes)
46,XY	Normal male karyotype with 46 chromosomes (an X and a Y chromosome)
47,XXY	Karyotype with 47 chromosomes (two X chromosomes, a Y chromosome)
47,XY,+21	Karyotype with 47 chromosomes (X and Y chromosomes); an additional chromosome 21 (trisomy 21)
13p	Short arm of chromosome 13
13q	Long arm of chromosome 13
13q14	Region 1, band 4 of the long arm of chromosome 13
13q14.2	Subband 2 of 13q14
2q-	Shortening of the long arm of chromosome 2
del(2)	Deletion in chromosome 2
del(2)(q21-qter)	Deletion in chromosome 2 of region 2, band 1 (2q21) of the long arm to the end (telomere) of the long arm (qter)
inv(4)	Inversion in chromosome 4
inv(4)(p11q21)	Inversion in chromosome 4, 4p11 to 4q21 (pericentric inversion)
dup(1)	Duplication in chromosome 1
inv dup(1)	Inverted duplication in chromosome 1
inv dup(2)(p23-p24)	Inverted duplication of the bands p23 to p24 in chromosome 2
r(13)	Ring-shaped chromosome 13 (deletion implied)
i(Xq)	Isochromosome for the long arm of an X chromosome
dic(Y)	Dicentric Y chromosome
idic(X)	Isodicentric X
t(2;5)	Reciprocal translocation between a chromosome 2 and a chromosome 5
t(2q-;5q+)	Reciprocal translocation between a chromosome 2 and a chromosome 5; the long arm of 2 has been shortened and that of 5 has become longer
t(2;5)(q21;q31)	Reciprocal translocation with the breakpoints in q21 of chromosome 2 and q31 of chromosome 5
der(2)	A ("derivative") chromosome derived from chromosome 2
t(13q14q)	Translocation of the centric fusion type, of the long arms of a chromosome 13 and a chromosome 14. It results in a single chromosome
ins(5)	Insertion into chromosome 5
ins(5;2)(p14;q22;q32)	The segment q22 to q32 of a chromosome 2 has been inserted into region p14 of a chromosome 5
fra((X)(q27.3)	X chromosome with a fragile site at position q27.3



A. Metaphase chromosomes of man (pairs 13 – 22, X and Y)

Preparation of Metaphase Chromosomes for Analysis

Since chromosomes are visible as individual structures under the light microscope only during metaphase (or under special conditions and for certain purposes, during prometaphase), every chromosome analysis requires dividing cells. In vivo, only bone marrow cells contain a sufficient proportion of cells in mitosis. Thus, in-vivo chromosome analysis of cells is limited to bone marrow. All other procedures for analyzing chromosomes in mitosis require culturing of suitable cells (cell culture). Most commonly, lymphocytes from blood are cultured for chromosome preparations.

Peripheral blood lymphocytes stimulated by phytohemagglutinin grow in a suspension culture. Their life span is limited to a few cell divisions. However, by exposing the culture to Epstein-Barr virus they can be transformed into a lymphoblastoid cell line with permanent growth potential. Such cultures are widely used because they are much easier to handle than adhesion cultures (p. 122).

In addition, fibroblasts from a piece of skin can be propagated in cell culture for analysis (see p. 122). However, since this procedure is somewhat elaborate and time-consuming, it is used only for certain purposes.

A. Chromosome analysis from blood

For the cell culture, either peripheral blood is used directly or lymphocytes are isolated from peripheral blood (T lymphocytes). A sample of about 2 ml of peripheral blood is needed. The blood is prevented from clotting by use of a heparinized syringe, since clumping of the blood cells precludes culturing (the proportion of heparin to blood is about 1 : 20). Peripheral blood or isolated lymphocytes are placed in a vessel with culture medium. The cells are generally stimulated with phytohemagglutinin, a protein from plants that unspecifically stimulates lymphocytes to divide. The culture requires about 72 hours at 37°C for cells to divide. Lymphocyte cultures are suspension cultures; i.e., the cells divide in culture medium without attaching to the culture vessel. Cell division is arrested and the culture is terminated by adding a suitable concentration of a colchicine derivative (colcemid) two hours prior to

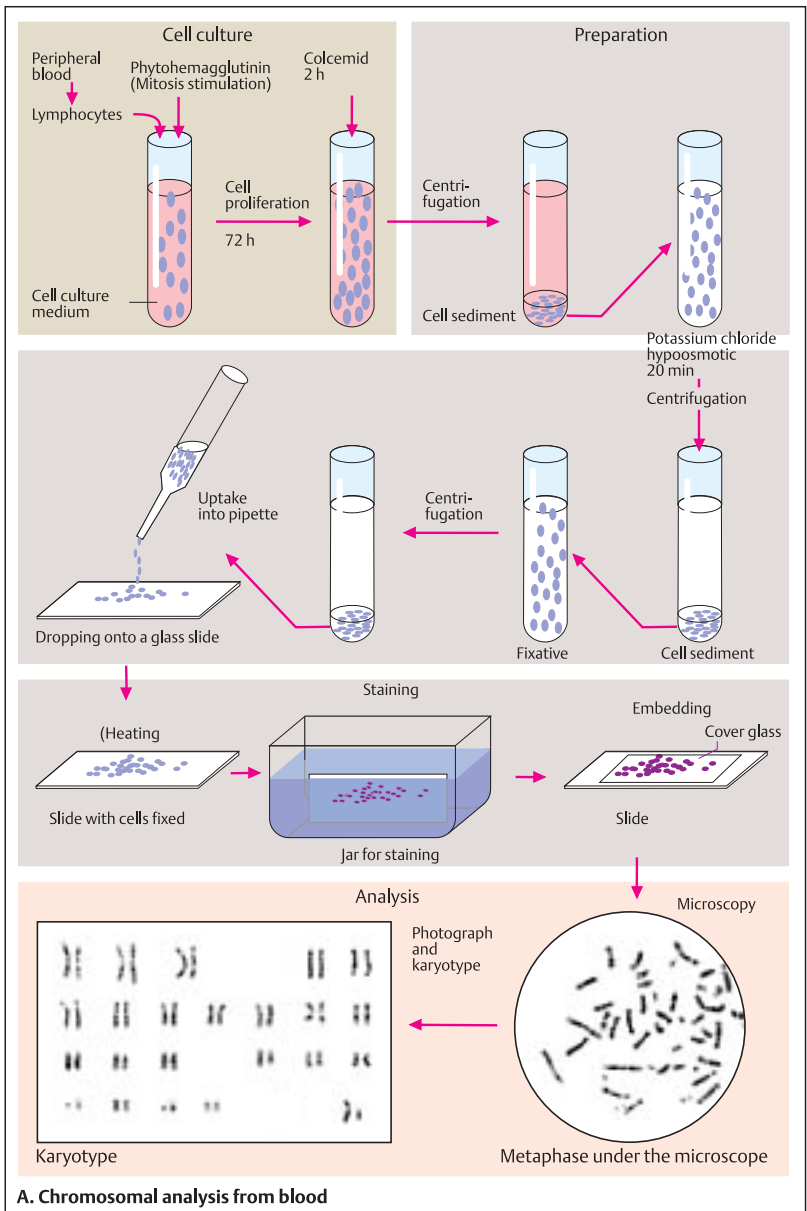
harvest. Colcemid interrupts mitosis during metaphase, so that a relative enrichment of cells in metaphase results.

Cell preparation is carried out as follows: the culture solution is centrifuged; the cell sediment is placed in a hypo-osmolar KCl solution (0.075 molar), incubated for about 20 minutes, and centrifuged again. The resulting cell sediment is placed in fixative. The fixing solution is a mixture of methyl alcohol and glacial acetic acid in a ratio of 3 : 1. Usually the fixative is changed two to three times with subsequent centrifugation. After that, the fixed cells are taken up in a pipette and dropped onto a slide. The preparation is stained, and the slide is covered with a cover glass.

At this point the cells are ready for analysis. Suitable metaphases are located under the microscope with about 100x magnification and are subsequently examined at about 1250x magnification. During direct analysis with the microscope, the number of chromosomes and the presence or absence of all chromosomes and recognizable chromosome segments are noted. Since the preparation procedure itself may induce deviations from the normal chromosome number or structure in some cells, more than one cell must be analyzed. Depending on the purpose of the analysis, between 5 and 100 metaphases (usually 10–15) are examined. Some of the metaphases are photographed with the microscope and can subsequently be cut out of the photograph (karyotyping). In this way a karyotype can be obtained from the photograph of a metaphase. The time needed for a chromosome analysis varies depending on the problem, but is usually 3–4 hours. Analysis and karyotyping time can be shortened by computer procedures.

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In Situ Hybridization in Metaphase and Interphase

In situ hybridization refers to procedures that demonstrate DNA sequences directly on chromosome preparations (in situ). Since resolution is relatively good (about 12×10^7 base pairs), the exact regional localization of a sequence on its corresponding chromosome can be determined.

A. Principle of in situ hybridization

Cells in metaphase or interphase are fixed on a slide and denatured to change the double-stranded DNA (1a) into single-stranded DNA (2). The metaphase or interphase preparation is then hybridized (3) with DNA sequences that are complementary to the region of interest and that have been labeled with biotin (1b). The hybridization site is made visible by means of a primary antibody against biotin; this antibody is bound to a fluorochrome (4), e. g., fluorescein isothiocyanate (FITC). Since the primary signal is quite weak, a secondary antibody (e.g., avidin) bound to biotin is attached (5). A further primary antibody can then be attached to the secondary antibody (6). This amplifies the signal, which can then be demonstrated by bright fluorescence under the light microscope.

B. Demonstration of the Philadelphia translocation in chronic myeloid leukemia (CML)

The Philadelphia translocation (1) in chronic myeloid leukemia (CML, see p. 332) can be demonstrated in metaphase (2) and in interphase (3) by means of in situ hybridization. When a probe for the *BCR* gene is used in interphase, the normal signal consists of two fluorescing dots, one dot on each chromosome 22. (On good preparations of metaphase chromosomes, one dot is seen over each chromatid and appears as a double dot on a chromosome.) When the probe includes the breakpoint of the translocation, three signals are visible: the largest over the normal chromosome 22, a small one over the *BCR* sequences remaining in the distal long arm of a chromosome 22 (22q), and another small one over the sequences translocated to the distal long arm of chromosome 9. (photographs by kind permission of T. Cremer, München; from Lengauer et al. 1992).

C. Translocation 4;8

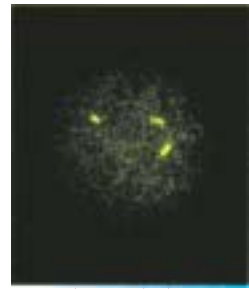
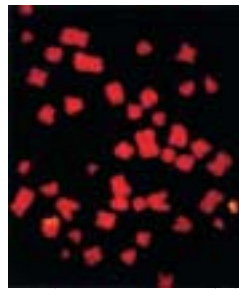
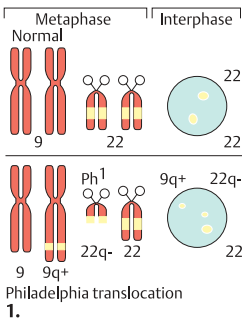
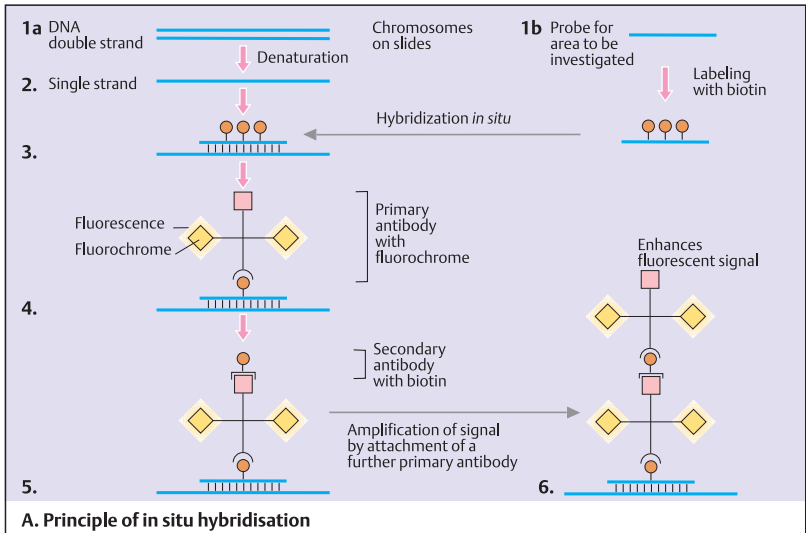
This preparation shows the translocation of part of the long arm of a chromosome 8 to the short arm of a chromosome 4 in a patient with Langer-Giedion syndrome. The hybridization was done with a 170-kb YAC (yeast artificial chromosome) that spans the breakpoint of the translocation in the 8q24 region. Three fluorescent signals result: over the normal chromosome 8, over the part of 8q24 translocated to chromosome 4, and over the sequences remaining on chromosome 8. Chromosomes 4 and 8 were hybridized with alphoid probes, which are specific for the centromere region. (Preparation by H. J. Lüdecke, Essen.)

D. Telomere sequences in metaphase chromosomes

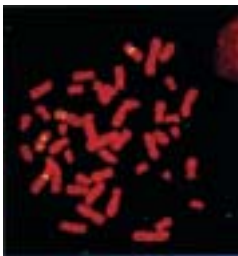
This illustration shows part of a human metaphase chromosome after in situ hybridization with telomere sequences (see p. 180). Each chromosome shows four signals, one over each end (telomere) of each chromatid, because the telomeric sequences are the same for all chromosomes. (Photographs by Robert M. Moyzes, 1991, Los Alamos, Laboratory, with kind permission of the author, *Scient. Amer.* August 1991, p. 34–41.)

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B. Demonstration of the Philadelphia translocation in chronic myelogenous leukemia



Specific Metaphase Chromosome Identification

The unambiguous identification of each chromosome is prerequisite to defining structural alterations associated with chromosomal imbalance. With banding pattern analysis, chromosomal resolution is limited by the size of the recognizable bands and the relative similarity of bands of different chromosomes. The smallest chromosomal region detectable by light microscopy in banded metaphase preparations is about 5–10 million base pairs (5–10 Mb). Such a segment could harbor 10–50 genes.

The methods utilize individual differences of DNA sequences of each chromosome and special techniques to induce different color images of each chromosome pair ("chromosome painting"). The DNA of metaphase chromosomes is first denatured (made single-stranded) and then hybridized on a slide *in situ* to DNA probes that make up a large collection of DNA fragments from one particular chromosome. Probes for all 24 human chromosomes are hybridized to the metaphase. Two approaches have proved particularly useful: multiplex fluorescence *in situ* hybridization (M-FISH, Speicher et al., 1996) and spectral karyotyping (SKY, Schröck et al., 1996). Other approaches and modifications exist; for example, the use of artificially extended DNA or chromatin fibers. In comparative genome hybridization (CGH), the quantitative difference in allelic sequences on homologous chromosomes is compared to detect deletions or duplications and amplifications.

A. Multiplex FISH

With this approach (M-FISH), sets of chromosome-specific DNA probes, each labeled with its own combination of DNA-binding fluorescent dyes, are hybridized to metaphase chromosomes. For each chromosome type, a specific multicolor bar code is constructed by using different YAC clones (yeast artificial chromosomes) containing the DNA probes to be applied. Only five different fluorophores are necessary for image analysis by epifluorescence microscopy using a charge-coupled device (CCD) camera to generate a composite image of each chromosome in a pseudocolor visualized by appropriate software. The karyotype

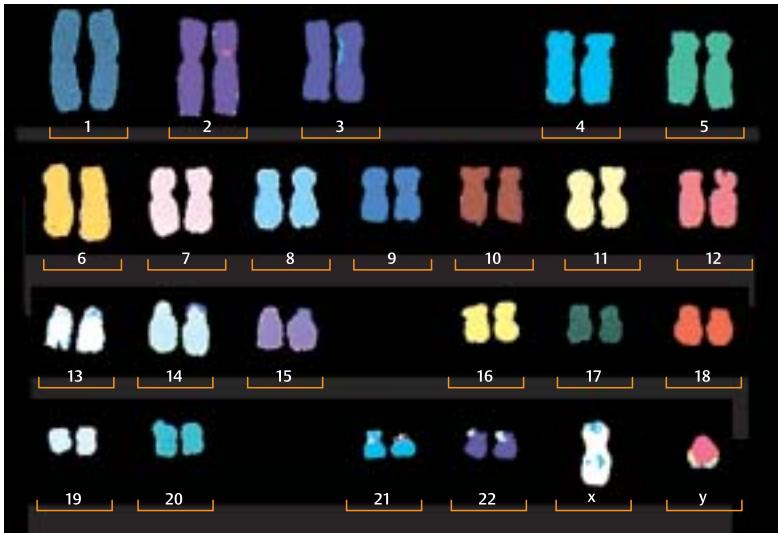
generated in this way is composed of the 22 pairs of autosomes and the X and the Y each in a different color. (Photograph courtesy of Drs. Sabine Uhrig and Michael Speicher, München).

B. Spectral karyotyping

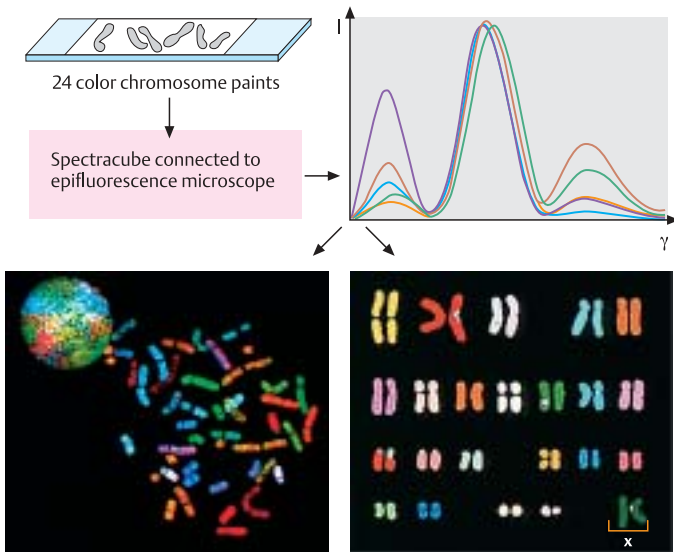
Spectral karyotyping (SKY) combines Fourier spectroscopy, CCD imaging, and optical microscopy. The emission spectra of all points in the sample are measured simultaneously in the visible and near-infrared spectral range. Twenty-four combinatorially labeled chromosome paint probes, one specific to each chromosome type, are hybridized to a metaphase after DNA denaturation. The emission spectra of the individual combinations of fluorophores are converted to a spectrum of different visible display colors by assigning blue, green, and red colors to specific spectral ranges of fluorescent wavelengths. The spectral karyotype is composed of a specific false color for each chromosome type. Spectral karyotyping has a wide range of diagnostic applications in the analysis of constitutional structural chromosome aberrations and cancer cytogenetics. (Photographs courtesy of Dr. Evelyn Schröck, Bethesda, Maryland).

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A . Multicolor fluorescence *in situ* hybridization karyogram



B. Spectral karyotyping

Numerical Chromosome Aberrations

Deviation from the normal chromosome number in a single pair of chromosomes is referred to as aneuploidy. In humans, numerical chromosome aberrations occur in about 1 in 400 newborns. An abnormality of the number of chromosomes occurs as a result of their abnormal distribution (nondisjunction) during meiosis I or II (meiotic nondisjunction). With meiotic nondisjunction, the aberration occurs in all cells of a resulting organism. Nondisjunction in meiosis I and meiosis II can be differentiated (see pp. 116). Abnormal chromosomal distribution during mitosis leads to an aberration in only a proportion of the cells (chromosomal mosaicism).

A. Triploidy

Triploidy refers to a deviation from the normal number of chromosomes in which each chromosome is present threefold instead of twofold. With tetraploidy, four copies of each chromosome are present. Triploidy arises when an abnormal oocyte with a double (46,XX) chromosome complement instead of a haploid complement (23,X) is formed. After fertilization by a normal spermatocyte, triploidy (69,XXX or 69,XXY) of maternal origin arises. In this case, two of the three complete sets of chromosomes are maternal. Triploidy may also arise as a result of abnormalities during spermatogenesis, resulting in an abnormal spermatozoon that does not contain the normal haploid chromosome complement, but rather the diploid (46,XY). In this case, the triploidy (69,XXY) is of paternal origin (see p. 402). A further cause of triploidy is dispermy, or fertilization of a normal egg by two normal sperm.

B. Aneuploidy

In a trisomy (1), only one of the chromosomes is present threefold; all other chromosome pairs are normal. Rarely, two different trisomies occur, for two different chromosomes (double aneuploidy). If one chromosome of a pair is missing, it is referred to as monosomy (2).

C. Origin of trisomy and monosomy

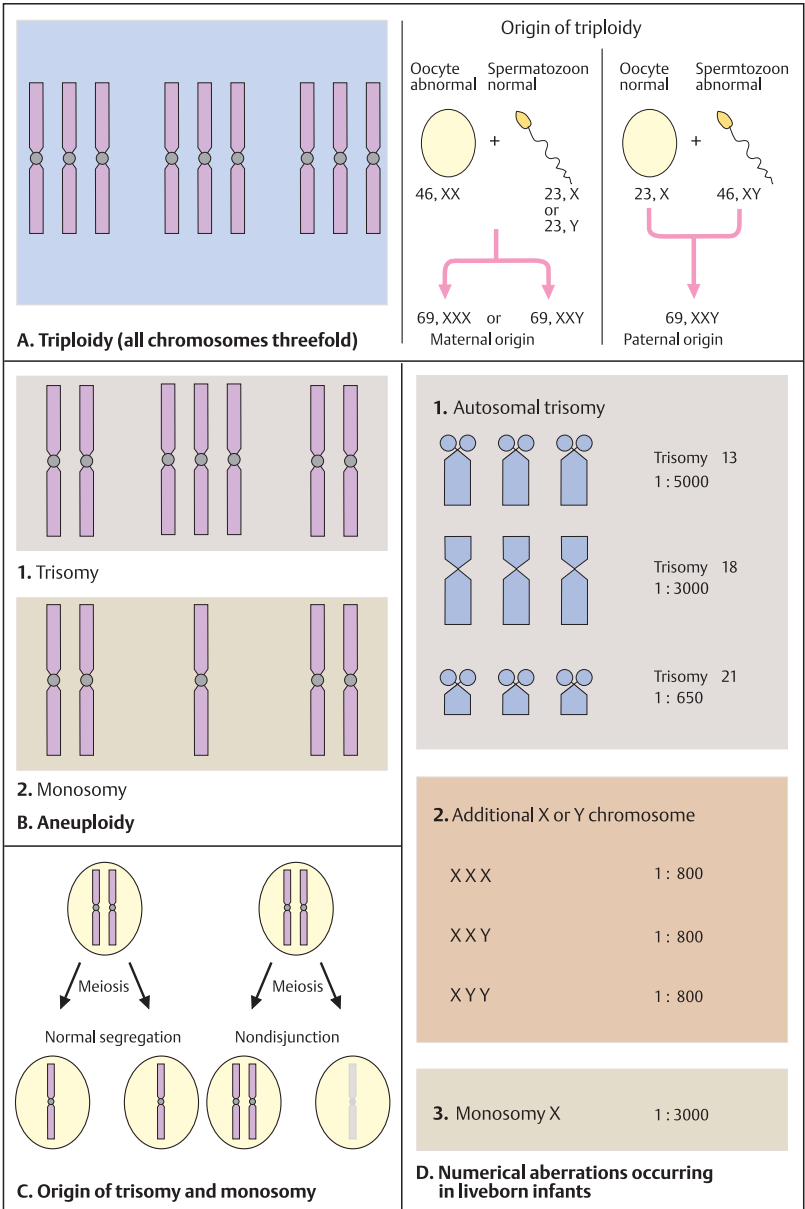
The result of normal meiosis (see p. 116), consisting of two cell divisions (not shown here), is a normal distribution and a haploid chromosome complement. With abnormal distribution (nondisjunction either in meiosis I or in meiosis II), one gamete is formed with an additional chromosome, whereas the other is missing a chromosome. After fertilization, the respective zygote contains either three copies of one chromosome (trisomy) or only a single chromosome of a pair (monosomy). Abnormal chromosome distribution can occur during oogenesis (maternal nondisjunction) or during spermatogenesis (paternal nondisjunction).

D. Abnormalities of chromosome number in humans

In humans, the following autosomal trisomies may occur in liveborn infants: trisomy 13 with a frequency of 1 in 5000 newborns; trisomy 18 with 1 in 3000; and trisomy 21 with about 1 in 650 newborns (1). Additional X or Y chromosomes occur in about 1 in 800 newborns, much more frequently than the autosomal trisomies (2). But unlike the autosomal trisomies, they usually do not lead to defined clinical pictures. Triple X (47,XXX) or an additional Y chromosome (47,XYY) are generally not clinically apparent. On the other hand, monosomy X (3) leads to the clinical picture of Turner syndrome (see p. 402), and XXY causes Klinefelter syndrome.

References

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Translocation

Translocation refers to an exchange of chromosome segments. A translocation can arise by centric fusion of two acrocentric chromosomes (Robertsonian translocation) or by exchange between two chromosomes (reciprocal translocation). With centric fusion, two complete chromosomes are involved; with reciprocal translocation, only a part of each of the two chromosomes is exchanged. In a translocation it is important to determine the breakpoints in each of the chromosomes involved.

A. Centric fusion of acrocentric chromosomes

Chromosome 14 and chromosome 21 (1) are the most frequently involved in fusions (about 1 in 1000 newborns). By fusion of the long arm of chromosome 21 (21q) and the long arm of chromosome 14 (14q), a chromosome t (14q21q) is formed (2). The satellite-carrying short arms of both chromosomes are lost, but this is insignificant. When germ cells (gametes) are formed, deviation from the normal chromosome number may result (3). Since chromosome 14 and chromosome 21 pair during meiosis, the following possible gametes may result: chromosome 14 alone (no chromosome 21), one chromosome 14 and one chromosome 21 (normal), the chromosome 14 fused to chromosome 21 (balanced), or the fused chromosome and one chromosome 21 (altogether two chromosomes 21).

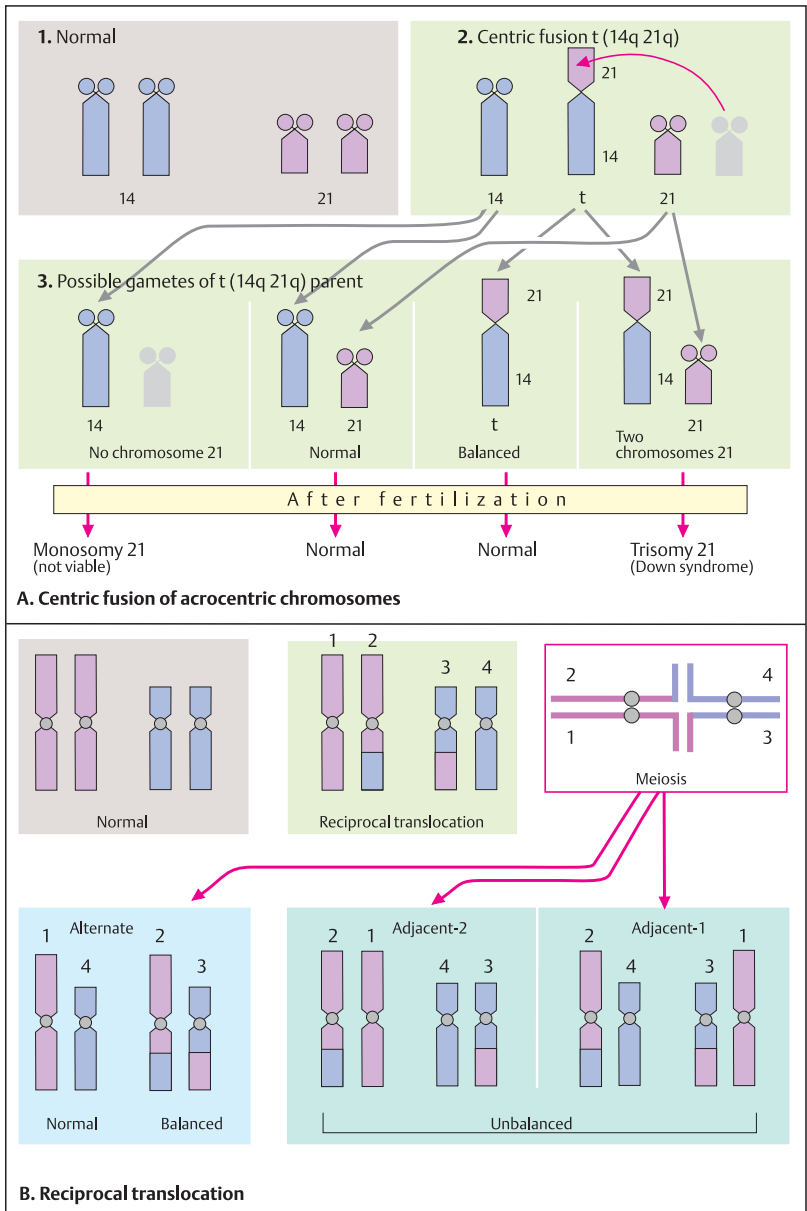
After fertilization, the corresponding zygotes contain either only one chromosome 21 (unviable monosomy 21), a normal chromosome complement, a balanced chromosome complement with the fused chromosome, or three chromosomes 21 (trisomy 21). In the latter case, the clinical disorder Down syndrome (formerly called mongolism) results (p. 400).

B. Reciprocal translocation

A reciprocal translocation is an exchange of chromosomal material between two chromosomes. Since usually no chromosomal material is lost or added with a reciprocal translocation, it does not cause clinical signs (i.e., it is balanced). However, carriers of a reciprocal translocation may form gametes with unbalanced chromosome complements. During meiosis, the chromosomes involved in the re-

ciprocal translocation take part as usual in the homologous pairing of meiosis I. Each of the chromosomes not involved in the translocation pairs with its homologous partner that is involved in the translocation. This leads to the formation of a characteristic quadriradial configuration of the involved chromosomes. When these four chromosomes separate (segregation) during anaphase of meiosis (see p. 116), one of three possibilities may occur: With alternate segregation, one gamete receives the two normal chromosomes, and the other gamete the chromosomes involved in the translocation, i.e., it is balanced.

With nonalternate segregation (neighboring or adjacent chromosomes), the two chromosomes on the left go into one gamete and the two chromosomes on the right into the other (adjacent-2). With the other possibility, the upper chromosomes go into one gamete, and the lower two into the other (adjacent-1). In each of the last two cases, an unbalanced distribution of the involved chromosome segments results. For example, after adjacent-2 segregation, gametes receive a partial duplication of the chromosome segment marked with red and a partial deficiency of the segment marked with blue (left pair of chromosomes) or a partial duplication of the blue segment and a partial deficiency of the red (duplication/deficiency). Different types of disorders result depending on the chromosome segments involved.



Different Types of Structural Chromosomal Aberrations

Structural changes in chromosomes can be classified according to cytological types and their effect on the phenotype. The main cytological types are translocation (exchange) (see p. 198), deletion (loss, see p. 182), inversion, insertion, isochromosome, dicentric chromosome, and ring chromosome (see below). According to their effects, they can be differentiated as balanced or unbalanced. With a balanced rearrangement, no chromosomal material has been lost or gained. In this case, there is no effect on the phenotype. In unbalanced aberrations, chromosomal material has either been added (partial duplication) or lost (partial deficiency). Simple types of structural aberration, such as deletion and duplication are shown on p. 182.

A. Inversion

An inversion is a 180-degree change in direction of a chromosomal segment. Prerequisite for every inversion is a break at two different sites, followed by reunion of the inverted segment. Depending on whether the centromere is involved, a pericentric inversion (when the centromere lies within the inverted segment) and a paracentric inversion can be differentiated.

B. The consequences of crossing-over in the inverted region

With homologous pairing during meiosis, an inversion loop is formed in the region of the inversion (1). When the inverted segment is relatively large, crossing-over may occur in this region (2). In the daughter cells, one chromosome may show a duplication (e. g., of segments A and B) and a deficiency (of segment F) (3), while the other chromosome shows deficiency of segments A and B and duplication of segment F (4). These chromosome segments are not balanced (aneusomy by recombination).

C. Isochromosome

An isochromosome arises when a normal chromosome (1) divides transversely instead of longitudinally, so that it is composed of two long arms (2) or of two short arms (3). In each case, the other arm is missing.

D. Dicentric chromosome

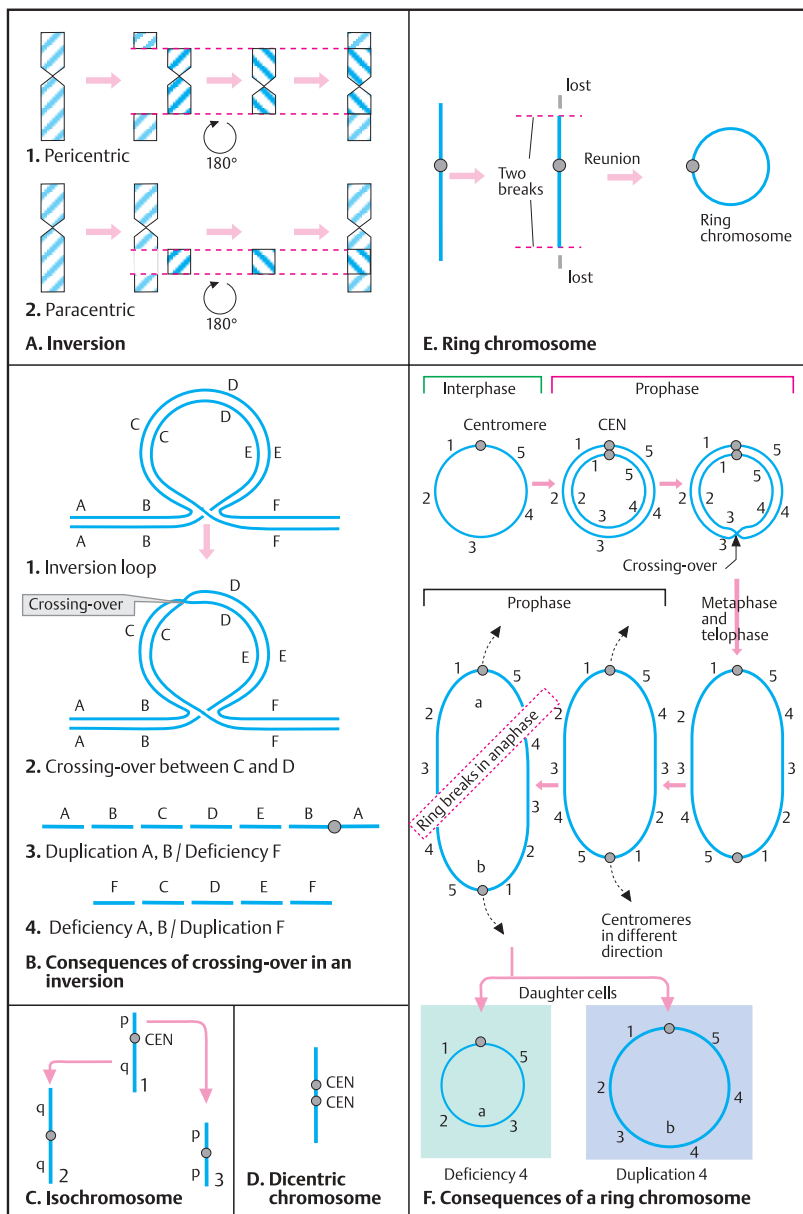
A dicentric chromosome contains two centromeres. It is unstable because it is usually torn apart during mitosis and its parts are divided between the two daughter cells.

E. Ring chromosome

A ring chromosome arises after two breaks followed by a joining of the two opposite ends. The distal segments are lost. Therefore, a ring chromosome is unbalanced.

F. Consequences of a ring chromosome

A ring chromosome is unstable because a break with reattachment ("crossing-over") of the chromatids during the prophase of mitosis usually leads to difficulties. In this case, a large ring-shaped chromosome with two centromeres arises during metaphase and telophase. Since the centromeres migrate in different directions during anaphase, the ring becomes disrupted. If this does not occur strictly symmetrically, two daughter cells will result with certain segments either missing (deficiency) or duplicated (duplication). In the example, one daughter cell with a deficiency of segment 4 and one daughter cell with a duplication of segment 4 are formed. Not infrequently, ring chromosomes are lost completely and a monosomy results.



Detection of Structural Chromosomal Aberrations by Molecular Methods

Structural chromosomal rearrangements occur with a frequency of about 0.7–2.4 per 1000 mentally retarded individuals. Small supernumerary chromosomes are observed about once in 2500 prenatal diagnoses. In both situations it is mandatory to identify the chromosome or chromosomes involved. Many small changes cannot be identified with conventional light-microscopic chromosome analysis, even in the best preparations using one of the banding techniques. The proportion of identifiable aberrations can be greatly enhanced by molecular cytogenetics (see p. 192) using fluorescence in situ hybridization (FISH).

A great variety of methods is available for identifying small rearrangements. Single-copy probes hybridizing to specific sites on individual chromosomes can be used to identify specific locations on a chromosome. Numerous probes from one chromosome can be applied to identify a whole chromosome (chromosome painting). In comparative genomic hybridization, the genomic DNA of a cell population, for example tumor cells, is hybridized to normal metaphase chromosomes. DNA segments that are overrepresented or underrepresented in the tumor tissue owing to duplication or deletion will appear as increased or decreased signals. The preparation of extended DNA fibers (fiber FISH) increases the resolution. The following are selected examples of the use of multiplex fluorescence in situ hybridization.

A. Derivative chromosome 1 with extra material: 46,XX,der(1)t(1:12)(q43;p13.3)

In the conventional G-banded karyogram (shown on the left) a small amount of additional chromosomal material is not visible. The karyogram by multiplex FISH on the right shows a small extra band at the end of one chromosome (1), shown by an arrow. The FISH analysis reveals that the extra band at the end of the long arm of a chromosome 1 (1q) is derived from a chromosome 12. The breakpoints could be determined to be in chromosome 1 in region 4, band 3 (1q43) and chromosome 12 in region 1, band 3.3 (12p13.3).

B. Additional isodicentric chromosome 15: 47,XY,+psu idic (15)(q11)

A small additional chromosome is present in the metaphase on the left (arrow) and in the corresponding karyotype on the right. The FISH karyotype reveals that the extra chromosome belongs to chromosome 15. It consists of a small isodicentric chromosome with a duplication of the proximal long arm 15q11 (region 1, band 1 of 15q).

C. Additional derivative chromosome 21

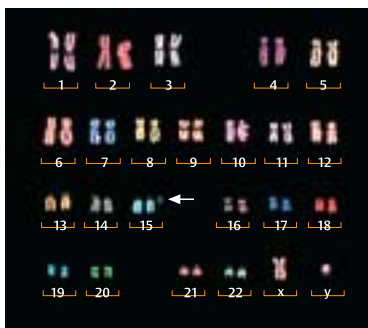
The metaphase on the left and the karyotype on the right show a small additional chromosome (arrow) with fluorescence in two colors. Detailed analysis of many metaphases revealed a compound chromosome consisting of material from a chromosome 18 and a centromere from a chromosome 21. Thus, the karyotype is unbalanced for additional material from a chromosome 18 (partial trisomy 18), which is a cause of developmental retardation. The origin of such a compound chromosome usually remains obscure. (All photographs kindly provided by Drs. Sabine Uhrig and Michael Speicher, München).

References

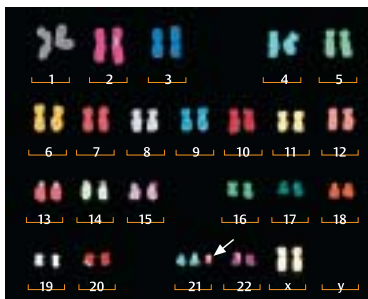
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A. Derivative chromosome 1 with extra material (arrow): 46,XX, der (1)t(1;12)(q43;p13.3)



B. Additional isodicentric chromosome 15 (arrow): 47,XY,+psu idic(15)(q11)



C. Additional derivative chromosome 21 (arrow): 47,XX,+der(21)t(18;21)(p11.2;q11.1)

Regulation and Expression of Genes

The Cell Nucleus and Ribosomal RNA

The cell nucleus is the main center from which cell functions are regulated. A highly important function of cells is to produce endogenous proteins (protein synthesis). The proteins, in turn, are required for innumerable vital processes, such as the catalyzation of complex biochemical reactions, production of energy, transport of molecules, etc. Cells from different tissues differ with respect to the genes that are expressed.

A. Cell nucleus and protein synthesis

Transcription and processing of the primary transcripts (RNA splicing) occur in the cell nucleus. RNA in the nucleus is bound to nuclear RNA-binding proteins for stabilization. The mature RNA is then released from the nucleus into the cytoplasm. For translation the mRNA must be bound to ribosomes. Ribosomes are complex protein structures made up of numerous subunits, which in turn are the products of individual genes (ribosomal genes).

B. Nucleolus and the synthesis of ribosomes

The nucleolus is a morphologically and functionally specific region in the cell nucleus in which ribosomes are synthesized. In man, the rRNA genes (200 copies per haploid genome) are transcribed by RNA polymerase I to form 45S rRNA molecules. After the 45S rRNA precursors have been produced, they are quickly packaged with ribosomal proteins (from the cytoplasm). Before they are transferred from the nucleus to the cytoplasm, they are cleaved to form three of the four rRNA subunits. These are released into the cytoplasm with the separately synthesized 5S subunit. Here they form functional ribosomes. The sizes of ribosomes, their subunits, and different types of ribosomal RNAs (rRNA) are given in Svedberg units (S). This is the rate at which a molecule sediments in a solvent. The S values are not additive. A functional ribosome consists of a small and a large subunit. In eukaryotes, the small

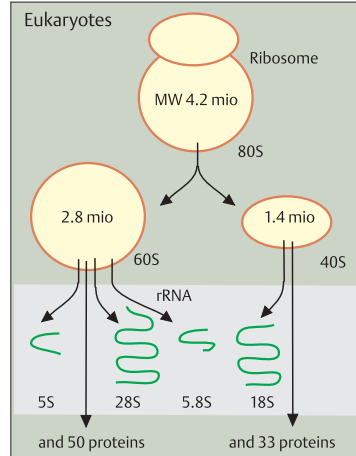
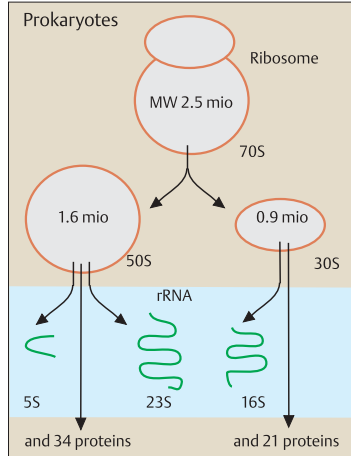
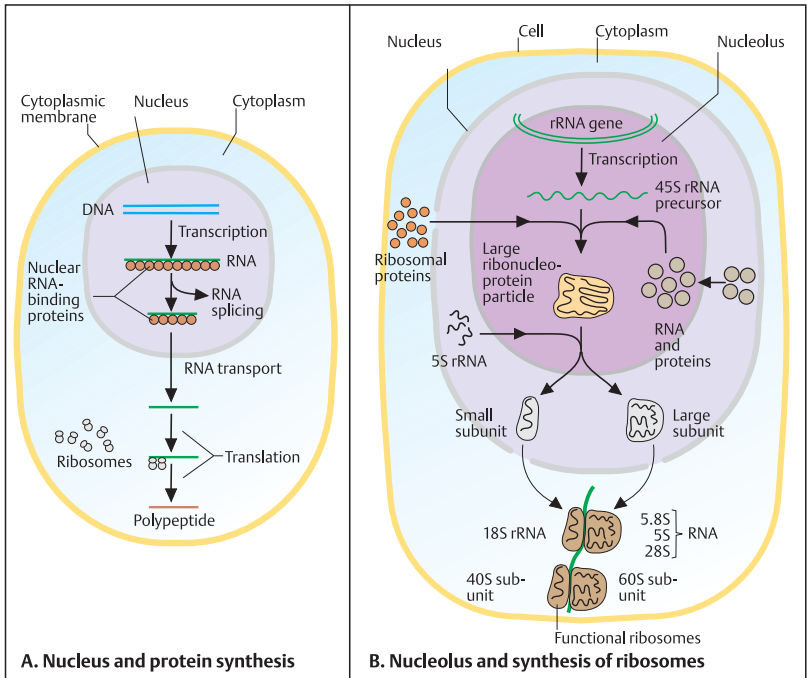
contains 18S rRNA, and the large subunit 5.8S, 5S, and 28S rRNA.

C. Overview of the structure and components of ribosomes

Ribosomes are the centers of protein synthesis. They provide the workplace and the tools. The 70S ribosome in prokaryotes consists of two subunits of 30S and 50S. The 50S subunit consists of a large (23S) and a small (5S) rRNA of ~2900 and 120 nucleotides, respectively, and 33–35 different proteins; the 30S subunit contains a large 16S rRNA and 21 proteins. The 50S subunit provides peptidyltransferase activity, while the 30S subunit is the site where genetic information is decoded. The 30S subunit also has a proofreading mechanism to minimize errors in translation. The whole ribosome has a molecular weight of 2.5 million daltons (MDa) and a sedimentation coefficient of 70S. The eukaryotic ribosome is much larger (4.2 MDa and 80S), with 60S and 40S subunits, which contain an array of rRNAs and proteins as shown in the figure. Recent observations of bacterial 30S and 50S ribosomal subunit structures at 5 Å resolution have helped to elucidate the details of ribosomal structure and function. (Figures adapted from Alberts et al., 1998).

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C. Overview of the structure and components of ribosomes

Transcription

The transcription of one strand of DNA into a complementary RNA molecule is the first step in gene expression. Multiple proteins (called transcription factors) form a transcription complex, which binds to DNA. Although there are differences in transcription in prokaryotes and eukaryotes, much of the basic process is the same. Transcription is catalyzed by RNA polymerase. RNA polymerase in *E. coli* has five subunits (two α , two β , and one sigma), each encoded by its own genes. RNA polymerases in eukaryotes are complex (see p. 214). Eukaryotic RNA polymerase consists of three different enzymes, which transcribe different types of genes.

A. Transcription by RNA polymerase

Transcription begins with recognition of a specific site by RNA polymerase (1). Here the double helix is opened and begins to unwind. RNA synthesis begins (initiation, 2) and continues with elongation (3). As the polymerase moves along the DNA, mRNA is synthesized. The DNA that has been transcribed rewinds into the double helix behind the polymerase. At termination (4), the RNA polymerase is removed from the DNA. At this point, the formation of the unstable primary transcript is completed. Since it is unstable, it is immediately translated in prokaryotes and modified (processed) in eukaryotes (see p. 50). All the processes are mediated by the complex interaction of a variety of enzymes.

B. Polymerase binding site

Bacterial RNA polymerase binds to a specific region of about 60 base pairs of the DNA. Several active centers can be identified (not shown here).

C. Promoter of transcription

Transcription must begin at a specific position of DNA, just upstream (at the 5' end) of a gene. This transcription initiation site is called a promoter. A promoter is a short nucleotide sequence of DNA that regulates the onset of transcription by binding to RNA polymerase. Two distinct promoter regions can be recognized above the transcription starting point. These sequences are evolutionarily highly conserved (consensus sequences). In prokaryotes, a

promoter with a consensus sequence consisting of six base pairs, TATAAT (also called a Pribnow box after its discoverer) is located 10 base pairs above the starting point; another region of conserved sequences, TTGACA, is located 35 base pairs above the gene (at the 5' end). These sequences are referred to as the -10 box and the -35 box respectively (the term "box" is derived from the sequence identity or similarity in all genes). In eukaryotes, the location and the sequences of the promoters differ slightly from those of the prokaryotes (see p. 212).

D. A transcription unit

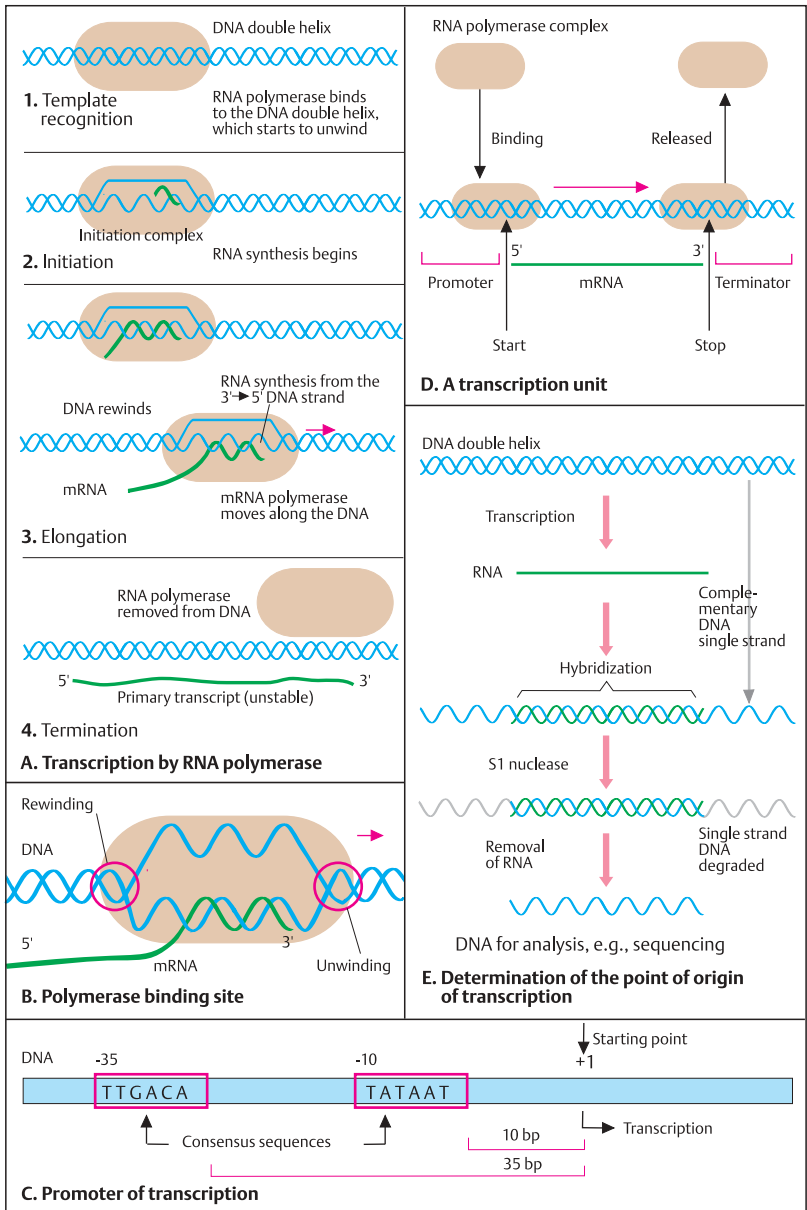
A transcription unit is all of the DNA sequences in a given segment that are used in transcription. It begins at the promoter and ends at the terminator. The region around the promoter at the 5' end is designated proximal; that around the terminator at the 3' end is designated distal.

E. Determination of the starting point of transcription

One way to identify an active gene is to determine the starting point of transcription. This can be done by comparing the RNA formed and the DNA template. After transcription, the RNA formed (single-stranded) is hybridized to a complementary single strand of DNA (RNA/DNA hybridization). An endonuclease (S1 nuclease) that cleaves only single-stranded DNA degrades the nonhybridized single strand of DNA, while the hybridized strand is protected (RNA protection assay). Subsequently, the RNA can be removed and the transcribed DNA segment analyzed (e.g., its size or sequence determined). (Figures after Singer and Berg, 1991).

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Control of Gene Expression in Bacteria by Induction

The regulation of gene expression is a basic function of prokaryotic and eukaryotic organisms. Prokaryotic organisms rely entirely on their ability to adapt rapidly to changes in external conditions. Substances usually present in the nutrient medium need not be synthesized by the bacterium itself. On the other hand, substances not present must be synthesized by the cell. The control of gene expression occurs at different levels. Regulator proteins may act as repressors (suppressing RNA polymerase activity) or as activators (inducing RNA polymerase activity). Control of prokaryotic genes is often facilitated in that functionally related genes usually lie together and therefore can be regulated together (operon) (see p. 234).

A. Induction of enzymes in bacteria

The presence of certain substances in the nutrient medium induces the synthesis of enzymes for their utilization in bacteria. An example in *Escherichia coli* (*E. coli*) is the activation of three enzymes for lactose catabolism by lactose. Within 10 minutes after lactose is added to the nutrient medium, the enzymes β -galactosidase, β -galactoside permease, and β -galactoside transacetylase increase manyfold (1). β -Galactosidase is the enzyme that splits lactose into galactose and glucose (2).

B. The lactose operon in *E. coli*

A series of genes whose regulation is coordinated is called an operon. Three structural genes that code for the synthesis of lactose-degrading enzymes (genes *lacZ*, *lacY*, and *lacA*) form the lactose operon (*lac* operon). These three genes are regulated by a promoter at the 5' end and are transcribed into a common mRNA (polycistronic transcript). Normally they show little activity because a *lac* repressor inhibits (*laci*) *lac* mRNA synthesis. The former is the gene product of the *lacI* regulator gene. *E. coli* can use lactose as its sole source of carbon and energy because large amounts of β -galactosidase can be synthesized within a short time.

C. Control of the *lac* operon

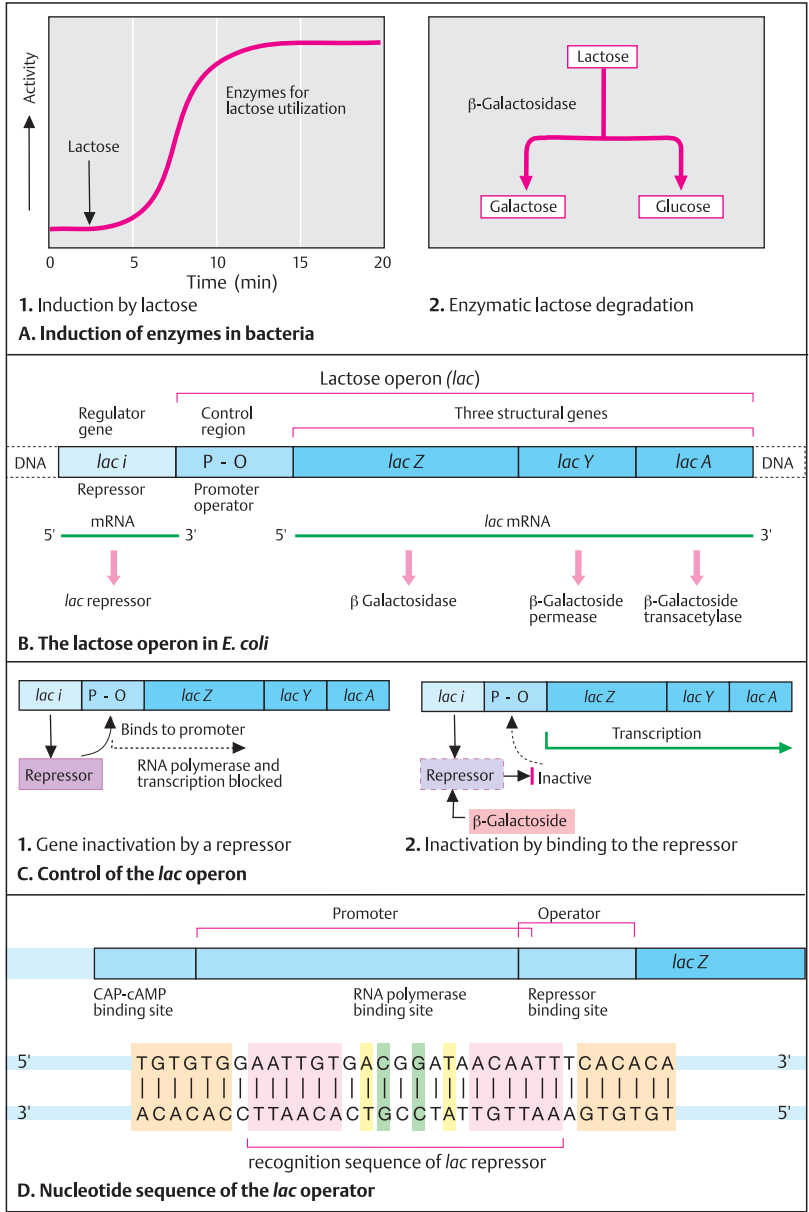
The three structural genes of the *lac* operon, *lacZ*, *lacY*, and *lacA*, are controlled by means of a repressor protein that binds to the promoter/operator region (P–O). When the repressor is bound to the P–O region, RNA polymerase cannot bind to the promoter region. Transcription is blocked (1), and the three gene products are not formed. The *lac* operon is activated when a β -galactoside molecule binds to one of the subunits of the repressor (2) and inactivates it. RNA polymerase can then bind to the promoter region and transcription can begin.

D. Gene-regulating nucleotide sequence of the *lac* operator

The activity of a gene is mediated by gene regulatory proteins that bind to DNA at specific sites. The repressor of the lactose operon (*lac* repressor) was the first such protein to be isolated (in 1966 by Gilbert and Müller-Hill). The repressor is a tetramer of identical 37-kDa subunits. Each has a binding site for the inducer. In the absence of the inducer, the repressor binds very tightly to the DNA in the operator region. The recognition sequence of the repressor is a short sequence of 28 nucleotide base pairs that are related by an axis of symmetry (shown as colored boxes). Symmetry matching is an important principle of protein–DNA interaction at a genetic switch. Gene-regulatory proteins can be distinguished by the specificity of the DNA sequences they recognize (compare the recognition sequences of restriction enzymes, p. 64).

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Control of Gene Expression in Bacteria by Repression

If a gene is usually expressed (is active), it is said to be constitutive. Gene expression in bacteria can vary considerably depending on the presence or absence of certain substances in the nutrient medium. An important mechanism for controlling transcription is a signal (termination signal) that can terminate transcription or translation. It lies between the promoter and the beginning of the first structural gene and is called the attenuator (attenuation of translation).

A. Regulation of synthesis of the amino acid tryptophan in *E. coli*

Tryptophan is an essential amino acid in eukaryotic organisms. Bacteria can synthesize tryptophan, but will do so only when it is not present in the nutrient medium (2). If tryptophan is added to the medium, enzyme activity for tryptophan biosynthesis decreases within about 10 minutes (1).

B. Biosynthesis of tryptophan by means of five enzymes and five genes

Tryptophan is synthesized from chorismate via four intermediates; this occurs in five steps, regulated by five enzymes. The enzymes are coded for by five genes (*trpA–E*) (CdRP = carboxyphenylamino-deoxyribulose-phosphate).

C. Tryptophan operon in *E. coli*

The tryptophan operon in *E. coli* consists of these five genes and their regulating sequences. The latter include promoter and operator, a leader sequence, and attenuator sequences. Translation of the five structural genes results from a continuous *trp* operon mRNA. In this, leader sequences coded for by L-sequence genes are connected in series. The attenuator sequences are part of the L-sequences. When tryptophan is present in the medium, translation of *trp* leader RNA is discontinued in the region of an attenuator sequence, before reaching the first structural gene.

D. The role of the attenuator

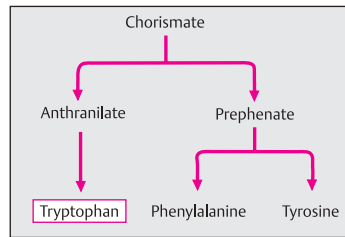
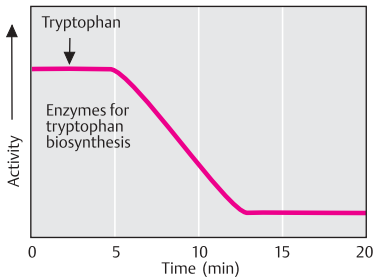
The weakening (attenuation) of the expression of the tryptophan operator is controlled by a sequence of about 100–140 base pairs (in the 3' direction) from the starting point of transcription (tryptophan mRNA leader). In the presence of tryptophan, the *trp* mRNA leader is interrupted in the region of an attenuator sequence (1), and translation does not take place. In the absence of tryptophan, translation is continued. The *trp* leader peptide contains two tryptophan residues (2). When tryptophan is deficient, translation is delayed and a stop signal weakened.

E. Attenuation of the *trp* operon

In *E. coli*, attenuation is often mediated by a tight relation of transcription and translation. The *trp* mRNA leader region can exist in two alternative base-pair conformations. One allows transcription, the other not. When tryptophan is present (1), ribosomes can synthesize the complete leader peptide. The ribosome closely follows the RNA polymerase transcribing the DNA template (not shown). The ribosome has passed region 1 and prevents complementary regions 2 and 3 from forming a hairpin by base pairing. Instead, part of complementary region 3 and region 4 form a stem and a loop, which favors termination of transcription. When tryptophan is deficient (2), the ribosome stalls at the two UGG *trp* codons owing to deficiency of tryptophanyl-tRNA. This alters the conformation of the mRNA so that regions 2 and 3 pair. The stem structure favoring termination is not formed, region 4 remains single-stranded, and transcription continues. (Figure adapted from Stryer, 1995).

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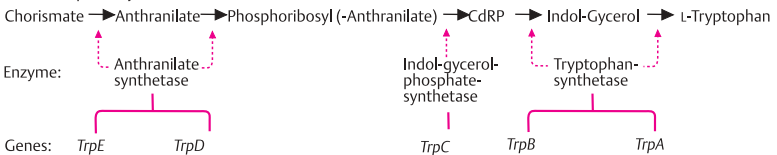
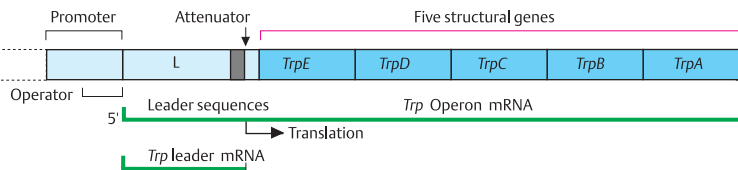
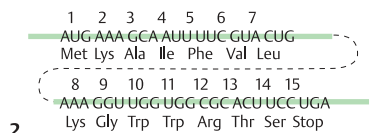
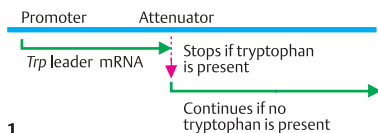


1. Decreased tryptophan synthesis

2. Synthesis of tryptophan

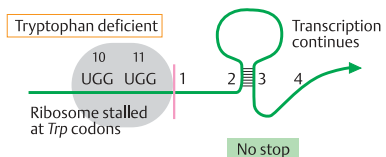
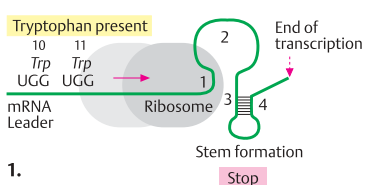
A. Influence on the synthesis of the amino acid tryptophan in *E. coli*

Metabolic pathway:

**B. Biosynthesis of tryptophan by five enzyme activities and five genes****C. Tryptophan operon in *E. coli***

1.

2.

D. The role of the attenuatorThe *Trp* leader peptide contains two *Trp* residues

1.

2.

E. Attenuation of the *trp* operon of *E. coli*

Control of Transcription

Transcription is controlled at promoters and other DNA sequences (enhancers) outside of the actual coding region. Transcription control in prokaryotes and eukaryotes differ in some respects and correspond in others. An important region for controlling gene expression, the promoter region, lies upstream (5' direction) of the coding sequence.

A. Promoter region

In prokaryotes, two important areas in the promoter region are 35 and 10 nucleotide base pairs upstream of (in the 5' direction) the starting point of transcription. Mutations in the region of the regulatory sequences (promoter region) in certain regions are extremely sensitive to base substitution (mutation), an indication of their importance. (Figure redrawn from Watson et al., 1987).

B. Assembly of general transcription factors to initiate transcription

The activation of polymerase II (Pol II) to transcribe most eukaryotic genes (1) requires an initiation complex assembled at the promoter. It consists of general transcription factors (TF) that associate in an ordered sequence. In the first step (2), TFIID (transcription factor D for polymerase II) binds to the TATA region. The TATA box is recognized by a small, 30-kDa TATA-binding protein (TBP), which is part of one of the many subunits of TFIID (the bending of the DNA by TBP is not shown here).

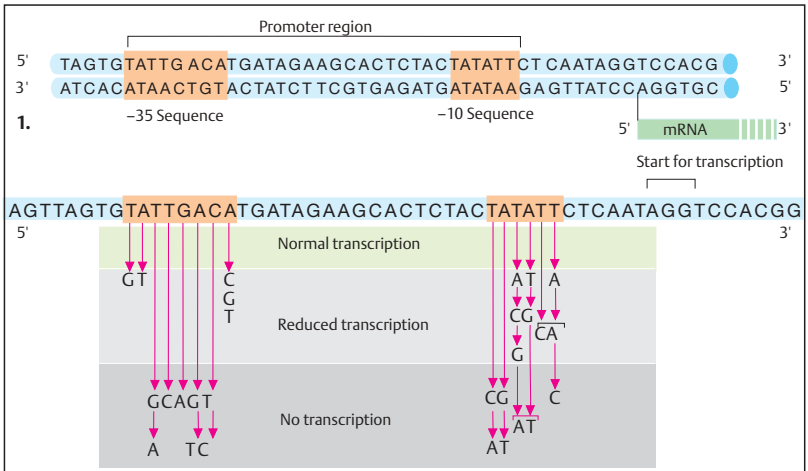
Following this, TFIIB can bind to the complex (3). Subsequently, other transcription factors (TFIIH, followed by TFIIIE) and Pol II escorted by TFIIF join the complex and assure that Pol II is attached to the promoter (4). The binding of TFIIIE extends the polymerase binding sites further downstream in the 3' direction. Pol II is then released from the complex and transcription can begin. A key step is phosphorylation of Pol II by a subunit of TFIIH, which is a protein kinase. Other activities of TFIIH involve a helicase and an ATPase. The site of phosphorylation is a polypeptide tail, composed in mammals of 52 repeats of the amino acid sequence YSPTSPS in which the serine (S) and threonine (T) side chains are phosphorylated. (The figure is a simplified scheme, redrawn from p. 421 in Alberts et al., 1994 and p. 631 in Lewin, 2000).

C. RNA polymerase promoters

Eukaryotic cells contain three RNA polymerases (Pol I, Pol II, and Pol III). Pol I is located in the nucleolus, synthesizes ribosomal RNA, and accounts for about 50–70% of the relative activity. Pol II and Pol III are located in the nucleoplasm (the part of the nucleus excluding the nucleolus). Pol II represents 20–40% of cellular activity. It is responsible for the synthesis of heterogeneous nuclear RNA (hnRNA), the precursor of mRNA. Pol III, responsible for the synthesis of tRNAs and other small RNAs, contributes only a minor activity of about 10%. Each of the large eukaryotic RNA polymerases (500 kDa or more) is more complex, with 8–14 subunits, than the single prokaryotic RNA polymerase. Each RNA polymerase uses a different type of promoter. RNA polymerase II cannot initiate transcription without a complex of general transcription factors (see B) that bind to a single upstream promoter (1). RNA polymerase I (2) has a bipartite promoter, one 170 to 180 bp upstream (5' direction) and another from about 45 bp upstream to 20 bp downstream (3' direction). The latter is called the core promoter. Pol I requires two ancillary factors, UBF1 and SL1. SL1 consists of four proteins including a TBP (see B) that cannot bind directly to the promoter. It binds to UBF1, after which Pol I can bind to the core promoter to initiate transcription (2). RNA polymerase III uses either upstream promoters or two internal promoters downstream of the transcription start site (3). Three transcription factors are required with internal promoters, TFIIA (a zinc finger protein, see p. 218), TFIIB (a TBP and two other proteins), and TFIIC (a large, more than 500 kDa protein).

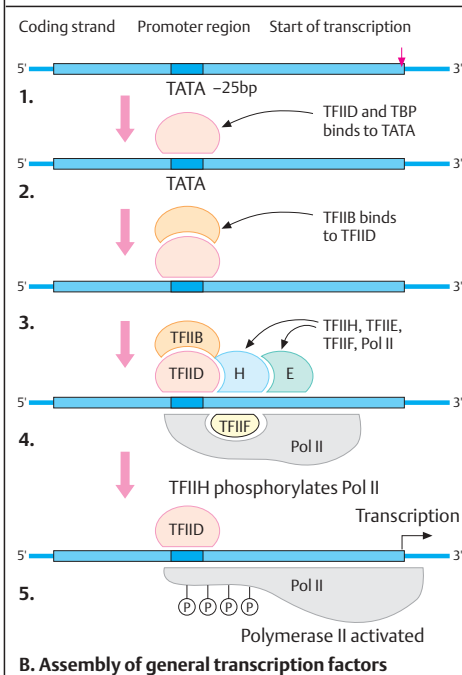
References

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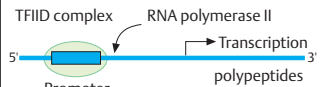


2. Effect of mutations in the promoter region on the rate of transcription

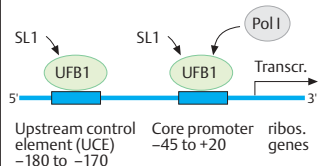
A. Promoter region in prokaryotes



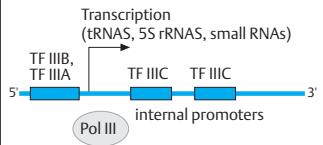
B. Assembly of general transcription factors



1. RNA polymerase II promoter



2. RNA polymerase I promoter is bipartite



3. RNA polymerase III promoters are upstream and downstream

C. RNA polymerase promoters

Transcription Control in Eukaryotes

Transcription in eukaryotes differs from that in prokaryotes in two main respects. In eukaryotes, one gene codes for a single polypeptide (monocistronic transcription unit) and the initial transcript is processed into mature messenger mRNA. This involves intron splicing (see p. 50) and substantial modification of the ends of the primary transcript.

A. Prototype of a eukaryotic structural gene

A structural gene is a gene that codes for a polypeptide gene product. It can be divided into sections involved in transcription (transcription unit) and regulatory sequences. Regulatory sequences are located both upstream (the 5' direction) and downstream (the 3' direction) of the gene. In addition, internal regulatory sequences may occur in introns. Some regulatory sequences are located far from the gene. Together with the promoter (see p. 206), they are required to regulate transcription.

B. Prototype of mature eukaryotic mRNA

Mature eukaryotic mRNA is produced from its precursor RNA by the removal of introns, addition of a 5' cap at the 5' end, and addition of numerous adenine nucleotides at the 3' end (polyadenylation). A noncoding sequence (5' leader) is located in front of the translation start signal (AUG), and a trailer sequence, at the 3' end in back of the translation stop signal (UAA). Both addition of the 5' cap and polyadenylation involve enzymatic reactions.

C. 7-Methyl-guanosine cap

The translation of eukaryotic mRNA is similar to that of prokaryotic mRNA, with two distinct differences: (1) transcription and translation occur at different locations in the eukaryotic cell: transcription occurs in the cell nucleus, and translation in the cytoplasm; (2) the 5' and 3' ends of eukaryotic mRNA have special structures. The structure at the 5' end is called a cap. Through the action of guanosine-7-methyltransferase, guanosine is bound by a triphosphate bridge to the first and second ribose groups of the precursor mRNA chain. The gua-

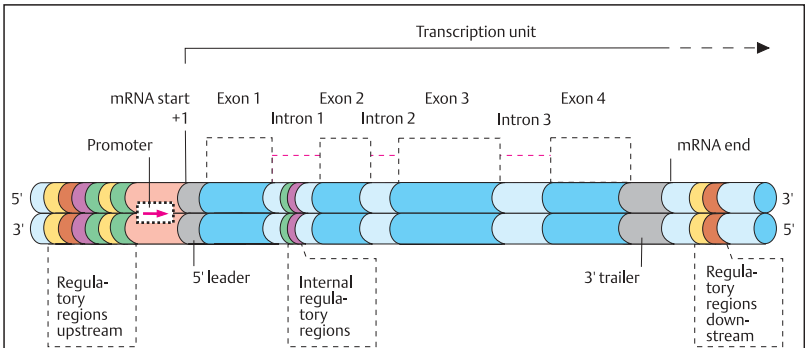
nosine is methylated in position 7, as are the two initial ribose residues at the beginning of the RNA chain. Except for the mRNAs transcribed by DNA viruses, eukaryotic mRNA usually contains a single protein-coding sequence (monocistronic messenger).

D. Polyadenylation at the 3' end

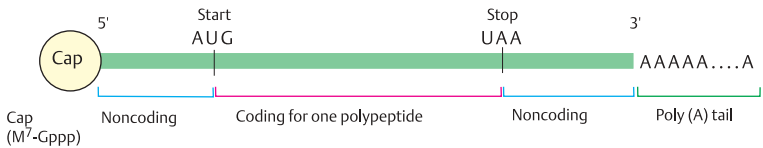
Eukaryotic termination signals have been less well recognized than the regulators of gene activity at the 5' end. Eukaryotic primary transcripts are split by a specific endonuclease shortly after the sequence AAAUAA. Subsequently, about 100 – 250 adenine nucleotides are attached to the 3' end of the transcript by means of a poly(A)-polymerase (polyadenylation). The poly(A) end binds to a protein. All mRNAs, except those that code for histone proteins, possess a poly(A) terminus. (Figures after Singer and Berg, 1991).

References

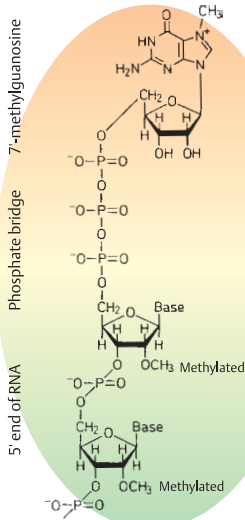
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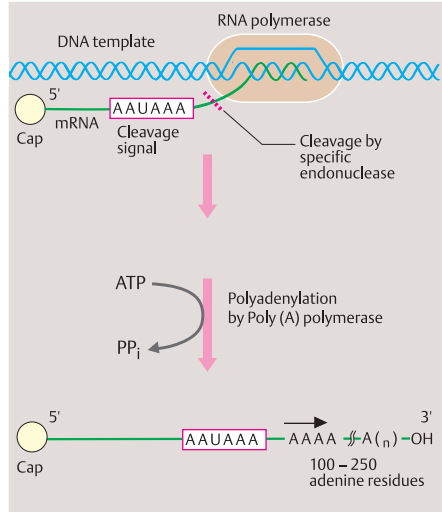
A. Prototype of a eukaryotic structural gene



B. Prototype of a mature eukaryotic mRNA



C. 7-Methyl guanosine cap



D. Polyadenylation at the 3' end

Regulation of Gene Expression in Eukaryotes

Precisely regulated gene expression is a prerequisite for producing and maintaining the many different types of cells and tissues of a multicellular organism. Cells differentiate into their particular cell types by means of combinations of expressed and repressed genes. During differentiation the tightly regulated genes function in the order, usually sequential, required for a particular cell fate (developmental pathways). Many regulator genes and their proteins have been identified (cf. part III, Genetics and Medicine). The following outlines some important principles of the specific control of gene expression in eukaryotic cells.

A. Levels of control of eukaryotic gene expression

In principle, expression can be regulated at four distinct levels. The first and by far the most important is primary control of transcription. Processing to mature RNA can be regulated at the level of the primary RNA transcript. A frequently observed process is alternative splicing (see D). Translation can be varied by RNA editing (see B for an example). At the protein level, posttranslational modifications can determine the activity of a protein. The cleavage of preproinsulin to form mature insulin, glycosylation or hydroxylation, and protein folding are some examples (see p. 32, 362).

B. RNA editing

RNA editing modifies genetic information at the RNA level. An important example is the apolipoprotein-B gene involved in lipid metabolism. It encodes a protein of 4538 amino acids, apolipoprotein B. This is synthesized in the liver and secreted into the blood, where it transports lipids. A related shorter form of the protein with 2153 amino acids, Apo B-48 (250 kDa, instead of 512 kDa for Apo B-100), is synthesized in the intestine. An intestinal deaminase converts a cytosine in codon 2158, CAA (glutamine), to uracil (UAA). This change results in a stop codon (UAA) and thereby terminates translation at this site.

C. Long-range gene activation by an enhancer

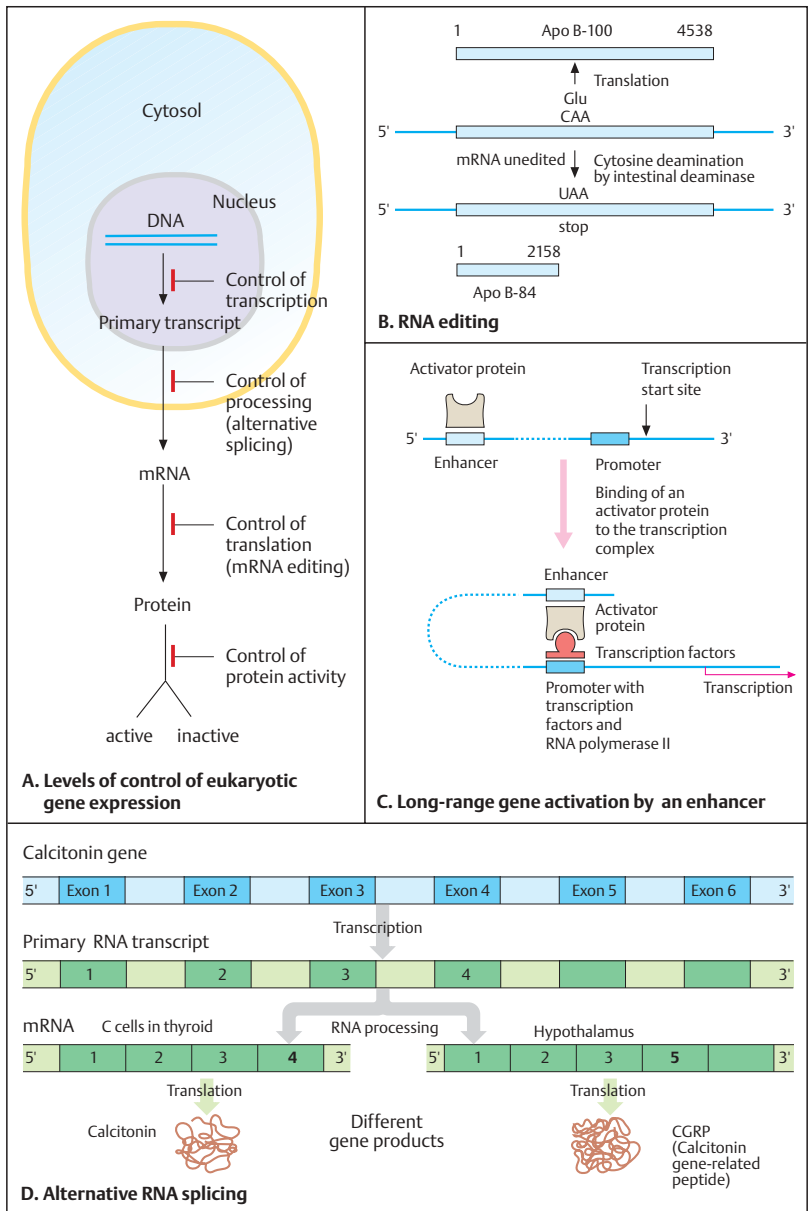
Enhancers control gene activity at a distance. An enhancer is a distant site involved in initiation of transcription (see p. 206). It may be located either upstream or downstream of the same DNA strand (*cis*-acting) or on a different DNA strand (*trans*-acting). Enhancer elements provide tissue-specific or time-dependent regulation. It is unclear how enhancers can exert their effect from a considerable distance. One model suggests that DNA forms a loop between enhancer and promoter. Activator proteins bound to the enhancer, e.g., a steroid hormone, could then come into contact with the general transcription factor complex at the promoter. Others might function as repressors (cf. transcription control in prokaryotes, p. 210).

D. Alternative RNA Splicing

A DNA segment can code for different forms of mRNA when different introns are removed from the primary transcript (alternative splicing). By means of alternative gene splicing, a gene can code for different, albeit similar gene products. This allows a high degree of functional flexibility. Numerous examples of differential RNA splicing are known for mammalian genes. For example, the primary transcript for the calcitonin gene contains six exons. They are spliced into two different types of mature mRNA. One, consisting of exons 1–4 (but not exons 5 and 6), is produced in the thyroid and codes for calcitonin. The other consists of exons 1, 2, 3, 5, and 6, but not exon 4. It codes for a calcitonin-like protein in the hypothalamus (calcitonin gene-related product, CGRP).

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DNA-Binding Proteins

Regulatory DNA sequences interact with proteins to exert proper functional control. Regulatory proteins can recognize specific DNA sequences because the surface of the proteins fits precisely onto the DNA surface. Three basic groups of regulatory DNA sequences can be distinguished: (1) sequences that establish the exact beginning of translation; (2) DNA segments that regulate the end, or termination; and (3) DNA sequences near the promoter that have specific effects on gene activity (repressors, activators, enhancers, and others).

A. Binding of a regulatory protein to DNA

Gene regulatory proteins can recognize DNA sequence information without having to open the hydrogen bonds within the helix. Each base pair represents a distinctive pattern of hydrogen bond donors (example shown in red) and hydrogen acceptors (example shown in green). These proteins recognize the major groove of DNA, where binding takes place. Here a single contact of an asparagine (Asn) of a gene-regulatory protein with a DNA base adenine (A) is shown. A typical area of surface-to-surface contact involves 10–20 such interactions. (Figure redrawn from Alberts et al., 1998, p. 276).

B. An α helix inserts into a major groove of operator DNA

One part of the protein, an α helix (the sequence-reading or recognition helix) is inserted into the major groove of DNA. Here the sequence Q-Q-Q-S-T (glutamine Q, serine S, threonine T) in the recognition sequence of the bacteriophage 434 repressor bonds with specific bases in a major groove of operator DNA. (Figure redrawn from Lodish et al., 2000, p. 351).

C. Zinc finger motif

Another group of proteins are called zinc fingers because they resemble fingers (see D). They are involved in important functions during embryonic development and differentiation. The basic zinc finger motif consists of a zinc atom connected to four amino acids of a polypeptide chain. Here, two histidine (H) and two cysteine (C) residues are shown in the schema on the left. The three-dimensional structure on the

right consists of an antiparallel β sheet (amino acids 1–10), an α helix (amino acids 12–24), and the zinc connection. Four amino acids, cysteines 3 and 6 and histidines 19 and 23, are bonded to the zinc atom and hold the carboxy (COOH) end of the α helix to one end of the β sheet. (Figure redrawn from Alberts et al., 1994, p. 411).

D. Zinc finger proteins bind to DNA

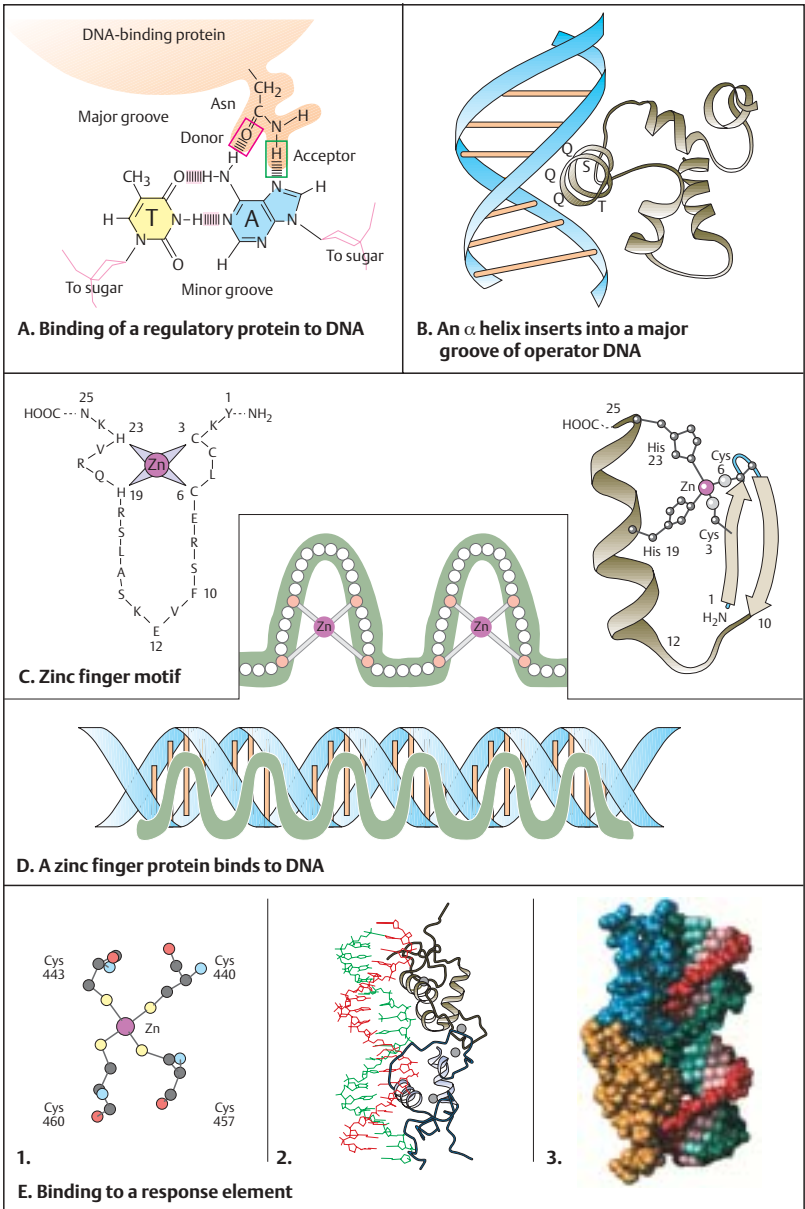
The interaction with DNA is strong and specific. Each protein recognizes a specific DNA sequence. As the number of zinc fingers can be varied, this type of DNA-binding has great evolutionary flexibility. (Figure redrawn from Alberts et al. 1994).

E. Binding to a response element

Many hormones and growth factors activate cell-surface receptors. In contrast, steroid hormones enter the target cells and interact there with a specific receptor protein in the cytosol. The hormone–receptor complex then migrates to specific sites of DNA. The hormone-binding domain will prevent binding to DNA unless the hormone is present. Activated receptors bind to specific DNA sequences called hormone response elements (HREs). Each polypeptide chain of the receptor contains a zinc atom bound to four cysteines (1). The skeletal model shows the two DNA-binding domains binding to the HRE in two adjacent major grooves of the target DNA (2). The space-filling model shows how tightly the recognition helix of each dimer of this protein fits into the major groove of DNA shown in red and green (3). (Figure redrawn from Stryer, 1995, p. 1002).

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Other Transcription Activators

Transcription activators are dimeric proteins with distinct functional domains: a DNA-binding domain and an activation domain. The DNA-binding domain interacts with specific regulatory DNA sequences. The activation domain interacts with other proteins that stimulate transcription. Transcription activators participate in the assembly of the initiation complex, for example, by stimulating the binding of transcription factor IID (TFIID, see p.212) to the promoter. Other activators may interact with general transcription factors. They provide a second level of transcriptional control.

A. Leucine zipper dimer

Most DNA-binding regulatory proteins recognize specific sites as dimers. One part of the molecule serves as the recognition molecule, the other stabilizes the structure. A particularly striking example is given by proteins with a leucine zipper motif. The name is derived from the basic structure. Two α helices are joined like a zipper by periodically repeated leucine residues located at the interface of the two helices. The two helices separate, form a Y-shaped structure, and extend into the major groove of the DNA (1). Leucine zipper proteins may be homodimers with identical subunits (2, 3) or heterodimers with different albeit similar subunits (4). The ability to form unlike dimers (heterodimerization) greatly expands the spectrum of specificities. The use of combinations of different proteins to control cellular functions is called combinatorial control. (Figure redrawn from Alberts et al., 1994).

A DNA-binding motif related to the leucine zipper is the helix–loop–helix (HLH) motif (not shown). The HLH motif consists of one short α helix and one longer α helix. The two α helices are connected by a flexible loop of protein.

B. Activation by steroid hormone binding

Transcriptional enhancers are regulatory regions of DNA that increase the rate of transcription. Their spacing and orientation vary relative to the starting point of transcription. An enhancer is activated by binding to a hormone–receptor complex. This activates the promoter, and transcription begins (active gene). Numerous important genes in mammalian develop-

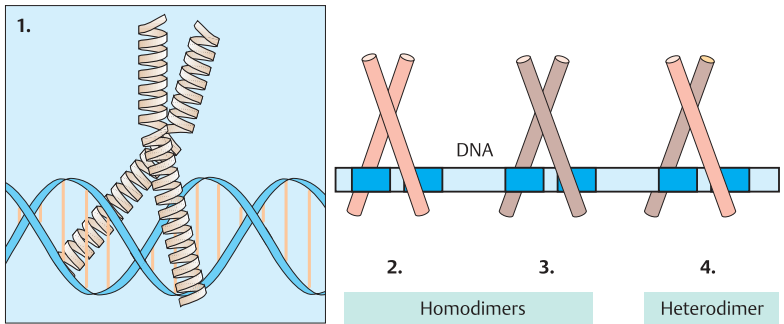
ment are regulated by steroids (steroid-responsive transcription). The latter include glucocorticoids and mineralocorticoids, the steroids of glycogen and mineral metabolism; sex hormones, which function in embryonic sex differentiation and control of reproduction; and others. Normal bone development and function are under the control of steroidlike vitamin D. Another steroidlike hormone is retinoic acid, an important regulator of differentiation during embryogenesis (morphogen). These hormones initiate their physiological effects by association with corresponding steroid-specific transcellular receptors (hormone–receptor complex).

C. Evidence of a protein-binding region in DNA

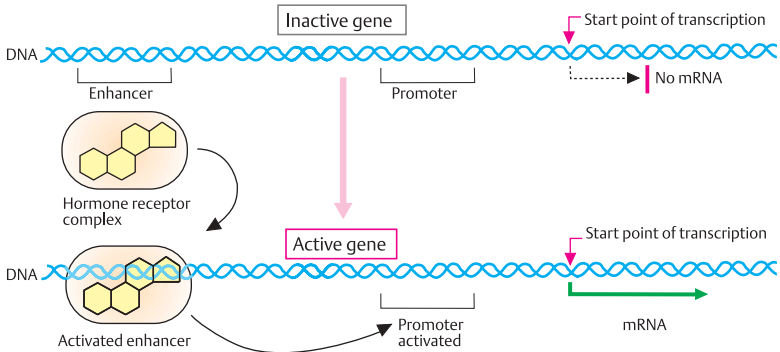
Protein-binding regions in DNA represent regulatory areas; thus, their analysis can yield some insights into gene regulation. Protein-binding DNA regions can be demonstrated in several ways. With band-shift analysis (1), protein-bound and non-protein-bound DNA fragments are differentiated using gel electrophoresis in direction towards the small fragments, a DNA fragment that is part of a DNA-protein complex migrates more slowly than a free DNA fragment of the same size. The DNA-protein complex is found at a different position ("band shift"). DNA footprinting (2) is another procedure for identifying protein-binding sites on DNA. The principle of DNA footprinting is that a protein-bound DNA region, e.g., the polymerase-promoter complex, is protected from the effects of a DNA-cleaving enzyme (DNAase I). Previously isolated DNA is cut into different fragments by DNAase, and the fragments are sorted according to size by gel electrophoresis. Since the DNA protein-binding region is protected from cleavage by DNAase I (DNAase I protection experiment), DNA bands from the binding region are missing ("footprint").

References

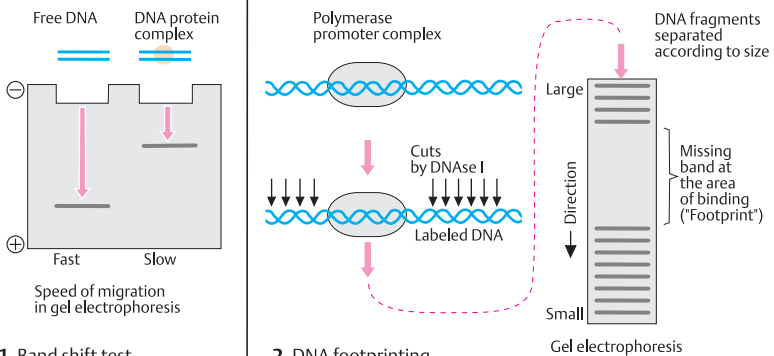
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A. Leucine zipper dimer



B. Activation by steroid hormone binding



C. Demonstration of a protein-binding site in DNA

Inhibitors of Transcription and Translation

A number of natural and artificial substances are able to inhibit transcription or translation. They can be used to treat cancer or as antibiotics to treat infections. Although most substances are unspecific and not suitable for therapeutic purposes, some are very specific and therefore important for the understanding of transcription and translation or for therapy. Basically, one can distinguish whether an agent interferes with transcription or with translation.

A. Insertion of actinomycin D between a GC base pair

Actinomycin D is a complex polypeptide produced by a species of streptomyces bacteria. It consists of a phenoxazone ring with two symmetrical side chains (1). It acts by becoming intercalated between two neighboring GC base pairs in double-stranded DNA. Viewed from the side (2), the inserted actinomycin D molecule is seen very distinctly within the DNA double helix. In the view from above (3) the actinomycin D molecule forms a narrow layer within the DNA double helix, bound by the two neighboring GC base pairs. The degree of inhibition by actinomycin D varies greatly. High concentrations of actinomycin D block replication, whereas low concentrations suffice to inhibit transcription.

B. Puromycin imitates an aminoacyl tRNA

Puromycin, a polypeptide from *Streptomyces alboniger*, blocks polypeptide synthesis in the ribosomes of prokaryotes and eukaryotes. Its action is based on the structural similarity with an aminoacyl tRNA. An aminoacyl tRNA is a tRNA molecule with an amino acid attached to its 3' end. Normally a peptide bond is formed by peptidyltransferase between the amino group of the incoming aminoacyl tRNA at the A (aminoacyl) position and the carboxyl group of the peptidyl tRNA at the P (peptidyl) position. The structure of puromycin resembles that of aminoacyl tRNA, but lacking an interaction with the codon it cannot be attached to the A position in the ribosome. The resulting polypeptidyl-puromycin adduct is removed from the ri-

bosome, and the protein synthesis ends prematurely.

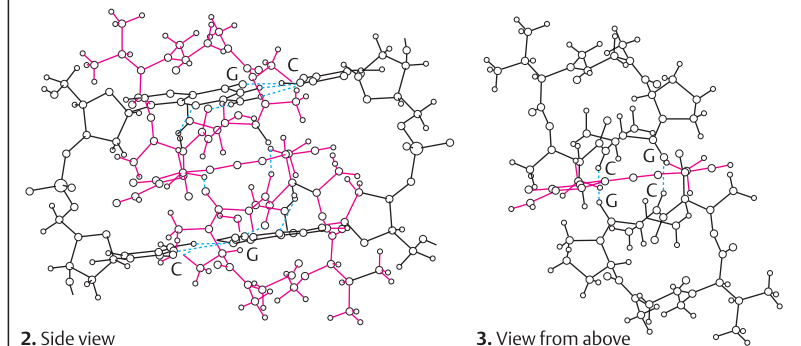
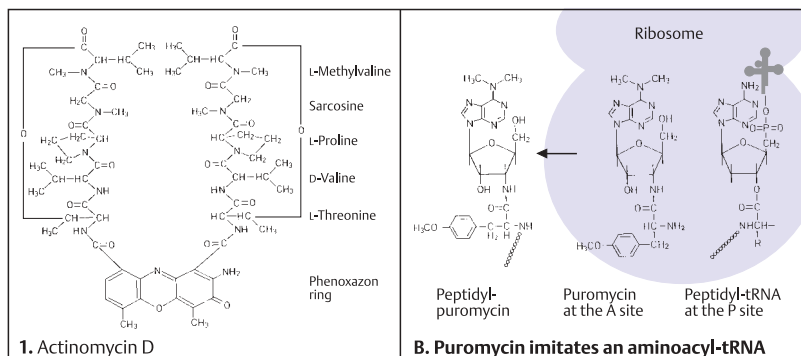
(Figures A and B from Singer and Berg, 1991).

C. Inhibitors of protein synthesis

Numerous naturally occurring and artificially produced substances inhibit protein synthesis by inhibiting transcription or certain phases of translation. Some have clinical significance as antibiotics; others are toxicologically significant. An example for the specificity of some inhibitors is α -amanitin, a dicyclic octapeptide of the fungus *Amanita phalloides*. In very low concentrations, it binds to RNA polymerase II and thereby blocks the formation of precursor mRNA in eukaryotes. In contrast, RNA polymerase I is insensitive to this toxin, and polymerase III binds to it only in higher concentrations. (Data after Singer and Berg, 1991).

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C. Inhibitors of protein synthesis